NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES, KARACHI

APPLIED HUMAN-COMPUTER INTERACTION

CLASS 5-B

Research-Based Project Report

AI-Driven Interaction & Intelligent Interfaces

Author
WARDAH AHMED
Roll number 24K-1045

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Note on AI use: Generative AI tools were used for drafting support and language editing. The study design, sessions and analysis are my own.

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Paper 1: How Does Conversation Length Impact User's Satisfaction?

Huang, Lin, He, Huang, and Huang. CHI EA '24, Honolulu, May 2024. Pennsylvania State University. arXiv:2404.17025.

1.1 Literature review

Conversational interfaces built on large language models have moved quickly from research prototypes to everyday tools, and a growing body of HCI work asks how design choices in these interfaces shape what users experience. One such choice is how long a conversation runs before the system gives its answer. A chatbot can respond at once, or it can ask clarifying questions first, and the two feel different to the person using them.

Huang et al. [1] tested this directly. They built two GPT-4 chatbots on Slack. One answered immediately, and the other asked 3, 5, or 7 follow-up questions before answering. Twenty-four participants put 40 questions taken from AskReddit to the bots, split into questions that invite conversation and questions that do not. For the conversational questions, self-reported satisfaction rose with conversation length, from an average of 3.64 with the immediate bot to 4.33 with the bot that asked seven follow-ups ($p = 0.041$). Crowd workers who rated the transcripts on MTurk saw no consistent quality gain. Helpfulness dipped at seven follow-ups and repetitiveness increased. Participants' own comments were mixed: some found longer exchanges more engaging, others found them repetitive, and their preferences depended on how urgent the question was and what kind of question it was. The authors conclude that a change in the form of a conversation does not necessarily mean a change in its quality [1].

Two points in this paper deserve a closer look. The first is that the headline claim is narrower than it first sounds. Satisfaction rose with length only for questions that invite conversation, and only the seven-follow-up condition reached significance. The result also comes from 24 people, so applying it to users in general is an inductive step: the conclusion is probable given what was observed, not guaranteed. The second point is the sample. The authors themselves note that participants were young and well educated, and as university students they were already comfortable with chatbots. Findings from that group say little about people who have never used one, yet a study framed as HCI work is meant to inform design for everyone.

Other work in this area points the same way: user differences change how a chatbot lands. Rahman and Desai [2] had 150 participants use a travel-planning chatbot at low, medium, or high personality intensity. The medium version was rated best on most measures, beating both extremes, and users whose own personality was closer to the agent's tended to rate it more favorably, so a behavior that suits one person can put off another. Bouyzourn and Birch [3] surveyed 115 UK students and interviewed four of them. Frequent use of ChatGPT went with higher trust, while a better understanding of how language models work went with lower trust, which means experience changes how people judge these systems. Abarca et al. [4] found a similar split in an explainability study with 10 novices and 10 experts. Novices rated the explanations useful and trustworthy, while experts were more critical of how complete they were. These studies show that novices and experienced users do not respond to the same interface in the same way. None of them tests clarifying-question behavior with novices.

1.2 Problem identification and research gap

Problem identification

Chatbots that ask clarifying questions are now common, and Huang et al. [1] suggest that the number of questions can raise satisfaction for some kinds of queries. The problem is that this evidence comes from a narrow group. A first-time user may not know why a chatbot is asking questions back. They may read a follow-up as the system failing to understand them, answer it wrongly, or stop partway through. A design that raised satisfaction for experienced students could then lower it, or block use altogether, for someone without that background. Control is a second problem. In the study the number of follow-ups was set by the researchers, so users had no say in how long the conversation ran, even though participants' comments show they wanted different lengths depending on urgency.

Research gap

Existing work on conversation length [1] tested young, educated users who already use chatbots and fixed the number of follow-up questions in advance. Related studies [2]-[4] show that user traits and experience change how people react to chatbots, but none examines how users with little or no chatbot experience respond to clarifying questions, or whether letting users control the number of questions helps them. This project addresses that gap.

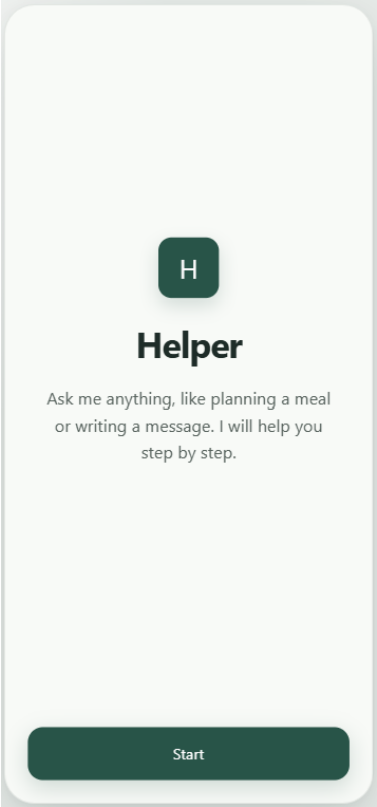
Research question

How do users with little chatbot experience respond to a chat interface in which they control the number of clarifying questions? The prototype in section 1.3 is built to explore this.

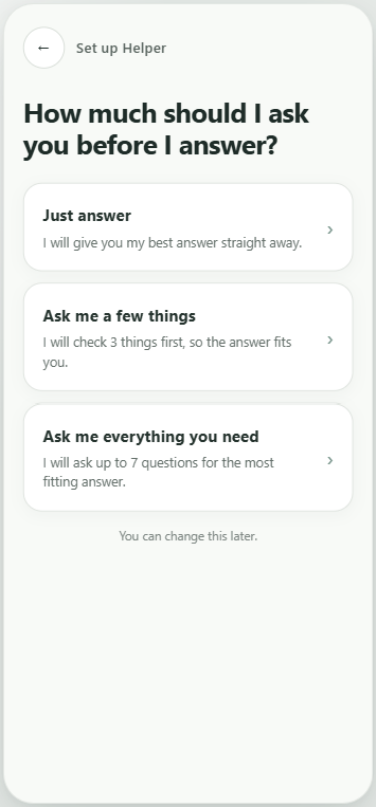
1.3 Prototype

The prototype is a clickable mobile app called Helper, built in Figma Make, a chatbot interface designed for someone who has never used one. It tests the idea from the gap in section 1.2: the user, not the system, decides how many clarifying questions the chatbot asks before it answers. Nothing is connected to a real model. The replies are fixed text, and the buttons link one screen to the next. The demo task is planning a dinner for guests, because that request naturally invites clarifying questions. The link opens the current version (version 2, described in section 1.4); version 1 is shown in the screenshots below. Prototype link: Figma Make prototype.

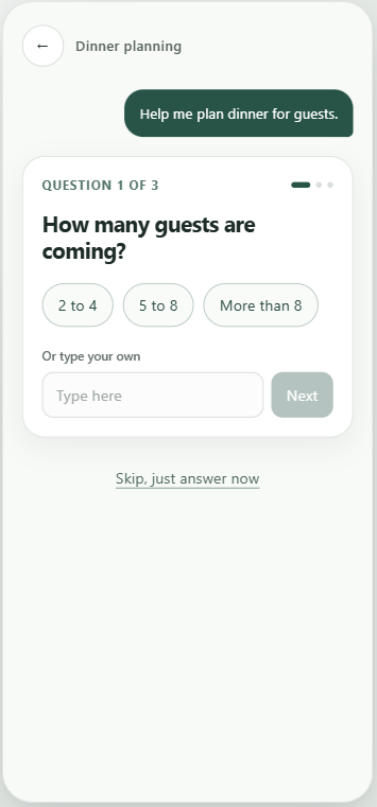
The setup screen offers three choices that map onto the conditions in the original study: “Just answer” is 0 questions, “Ask me a few things” is 3, and “Ask me everything you need” is up to 7. This keeps the prototype comparable with the paper while giving the choice to the user. All text is plain English with no terms such as model or prompt.



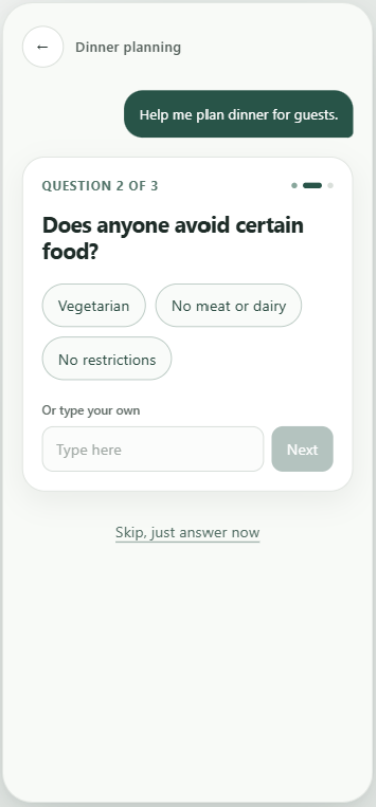
Screen 1: Welcome



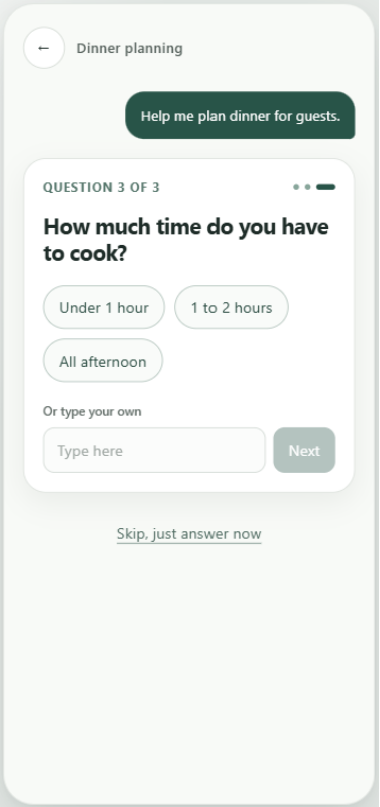
Screen 2: Setup. The user picks how much Helper asks



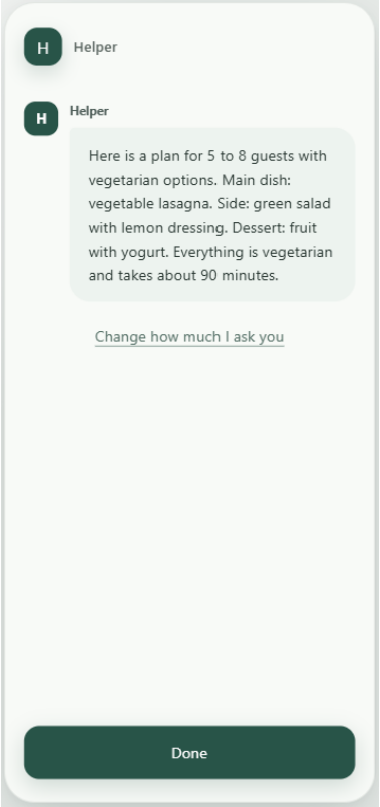
Screen 3: Question 1 of 3, with quick-reply chips and a Skip link



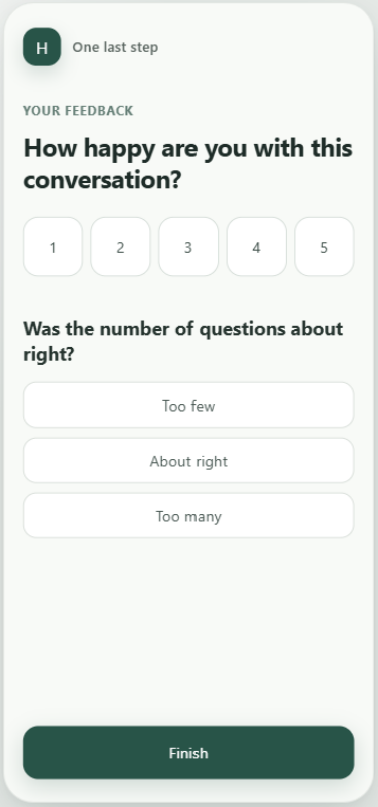
Screen 3: Question 2 of 3



Screen 3: Question 3 of 3



Screen 4: Answer, with a link to change the setting



Screen 5: Feedback, the data collected in section 1.4

Design choices tied to the gap

Quick-reply chips mean a novice never has to type. The visible counter and the Skip link give the user control over a process the original study imposed. The “Change how much I ask you” link on the answer screen lets people correct their first choice once they see how it feels. Every label is a full sentence a first-time user can read without help.

1.4 User study and evaluation

The prototype was evaluated in a small, qualitative usability study with three participants. The aim was to find out whether people with little chatbot experience could understand the setup choices, noticed the controls, and felt the number of questions suited them, and to collect problems in the design. Three people cannot support statistical claims, so the findings below are observations, not results that generalise.

Procedure

Each participant used the prototype on a phone, one at a time, with the researcher present. They were told to imagine planning a dinner for guests and to use the app as they normally would, speaking their thoughts aloud. The researcher did not explain the settings and helped only when the participant was stuck. Afterwards the participant answered the two questions on the feedback screen (satisfaction from 1 to 5, and whether the number of questions was too few, about right or too many) and gave short spoken comments. The researcher recorded which setup option was chosen, whether the Skip link was used, and whether the “Change how much I ask you” link was noticed.

Table 1. Observations from the three sessions.

ID	Age	Option chosen	Used Skip	Noticed change link	Satisfaction (1 to 5)	Number of questions
P1	16	Ask me a few things (3)	No	No	4	Too few
P2	23	Ask me a few things (3)	No	No	3	About right
P3	47	Just answer (0)	No questions shown	Not recorded	1	About right

Findings

Choice of setting. Two participants chose “Ask me a few things” and one chose “Just answer”. Nobody chose “Ask me everything you need”, so that option was not tested. Satisfaction was 4, 3 and 1. P1 found three questions too few and wanted more, in particular questions about culture and ethnicity, which suits a dinner for guests. P2 first thought the questions were too few, then too many, then realised there were only three and said it did not matter. This is a small sign that people differ on how much they want to be asked, which is the point of letting the user decide, but it is not evidence that it improves satisfaction.

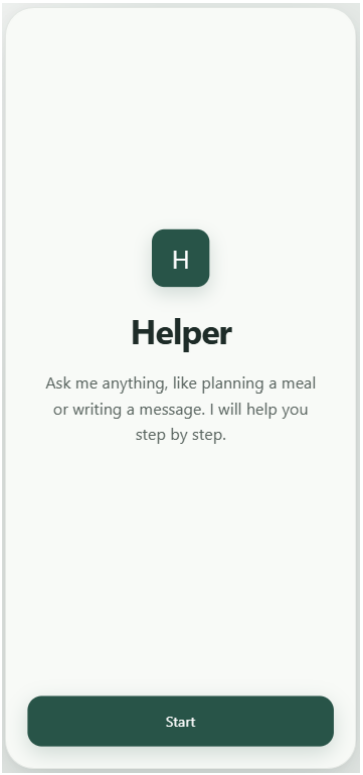
The answer was mistaken for a summary. Both P1 and P2 did not see the meal plan as a plan. P2 took it for a summary of what she had clicked through, and P1 pressed Done without registering it. P2 suggested that the message should start below the “Helper” label so it is clear it comes from the app. P3, who skipped the questions, did see the plan.

Controls were missed or unused. None of the participants used the Skip link. Because the flow had only three questions, this may mean Skip was not needed, not that it went unnoticed. P1 and P2 did not notice the “Change how much I ask you” link. P1 also pointed out that there was no way to go back after pressing Done.

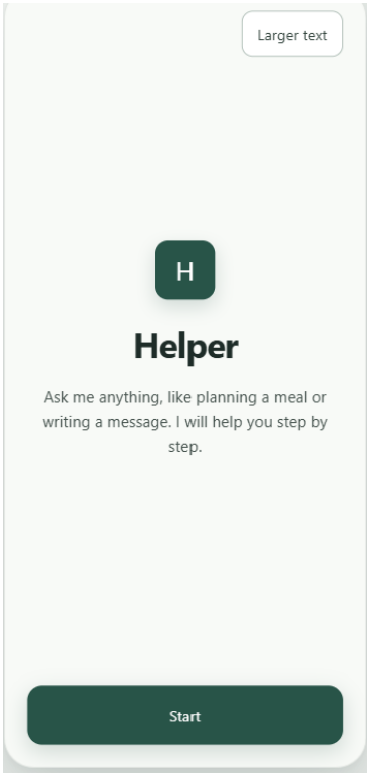
Legibility. P3 had difficulty reading the small on-screen text, and the researcher read the options aloud to her, so the session did not follow the plan of helping only when stuck. Her satisfaction rating of 1 reflects the difficulty of reading the screen and should not be read as a response to the question-length setting, since she skipped the questions. It does show that the small, light grey helper text under each option is hard to read for some users. P1 said she could use something like this.

Changes made in version 2

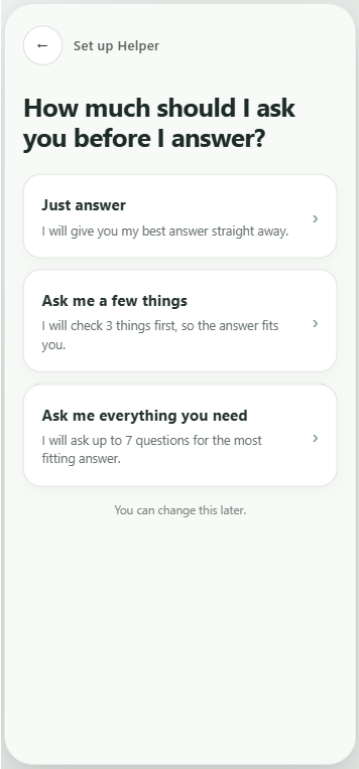
The prototype was revised after the three sessions. The text size and contrast were increased on every screen, and a “Larger text” button was added to the welcome screen, which enlarges the text on the screens that follow. On the answer screen, the message now sits under a “Helper” label and shows the meal as a card headed “Your dinner plan” with the main dish, side and dessert on separate lines, so it can no longer be mistaken for a summary. The small change link became a visible outlined button. The feedback screen gained a back arrow, so there is a way back after Done, and a closing “Thank you” screen with a “Start again” button now follows Finish. A culture question (“Is there a food tradition or culture you would like the dinner to follow?”) was added to the question flow, and the largest setting now reads “up to 4 questions”, while “Ask me a few things” still shows 3. Version 2 has not yet been tested with participants.



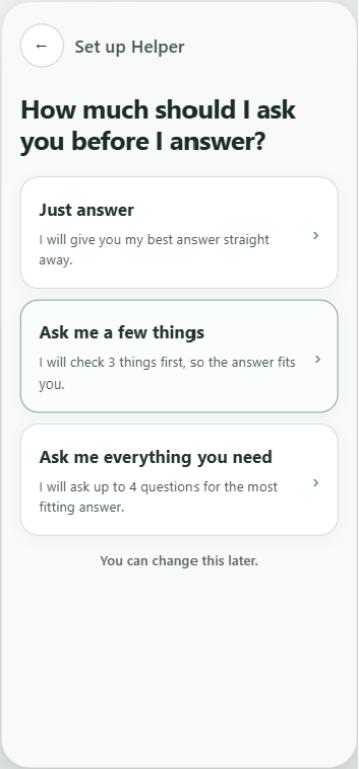
Welcome, version 1



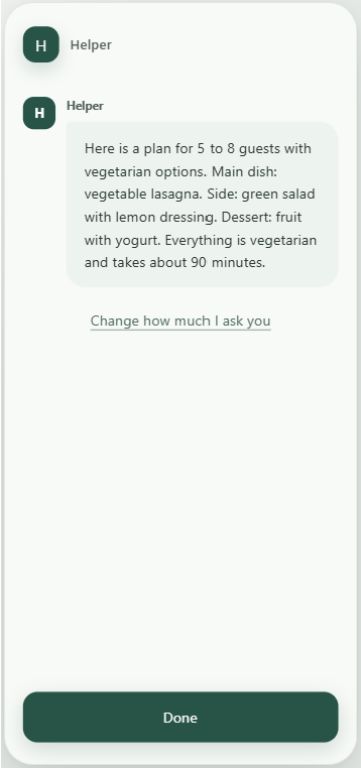
Welcome, version 2: larger text and a Larger text button



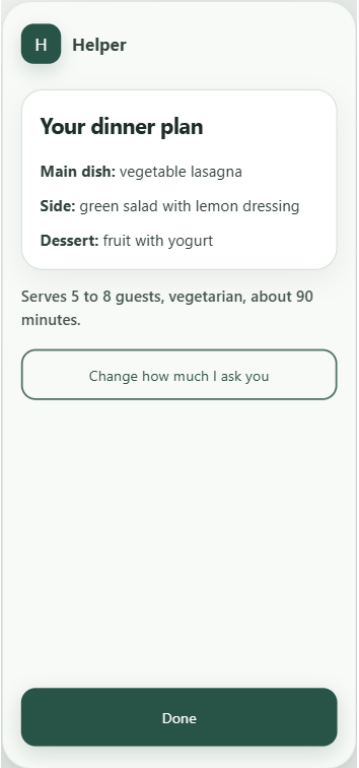
Setup, version 1



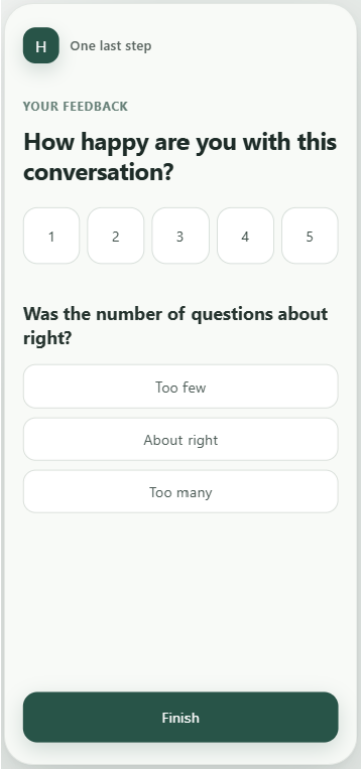
Setup, version 2: larger, darker text



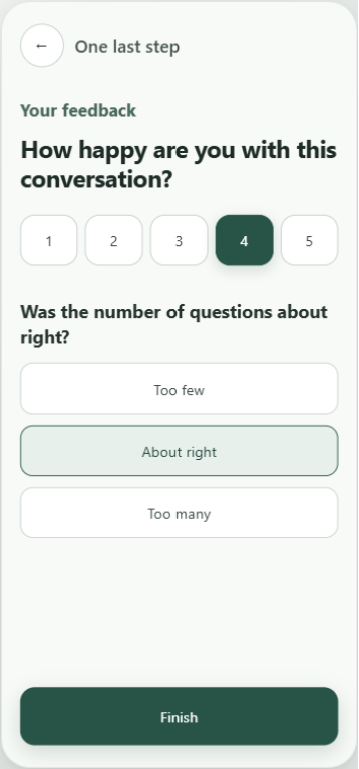
Answer, version 1



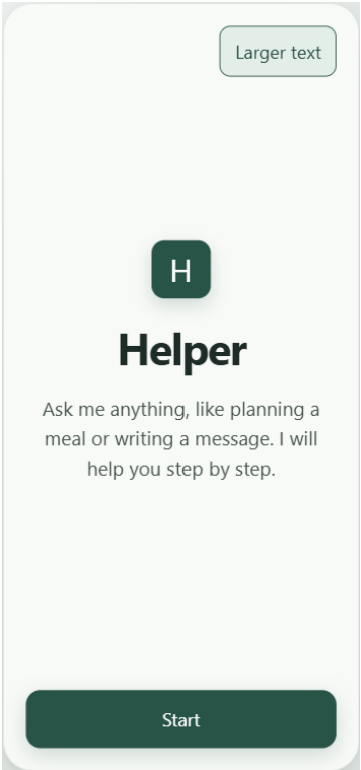
Answer, version 2: Helper label, plan card, visible change button



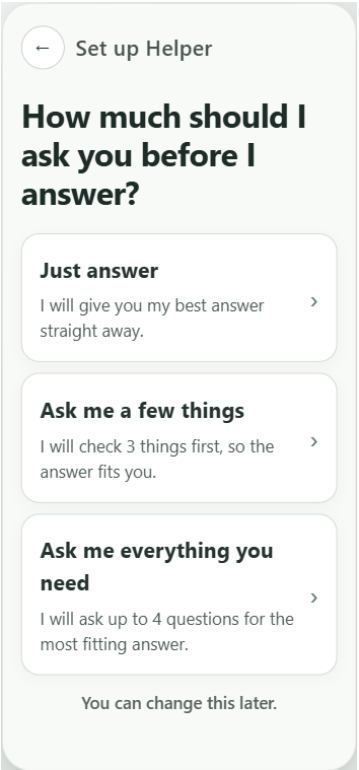
Feedback, version 1



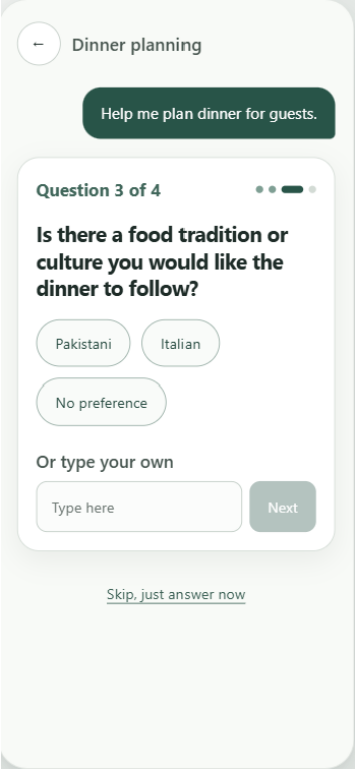
Feedback, version 2: back arrow added



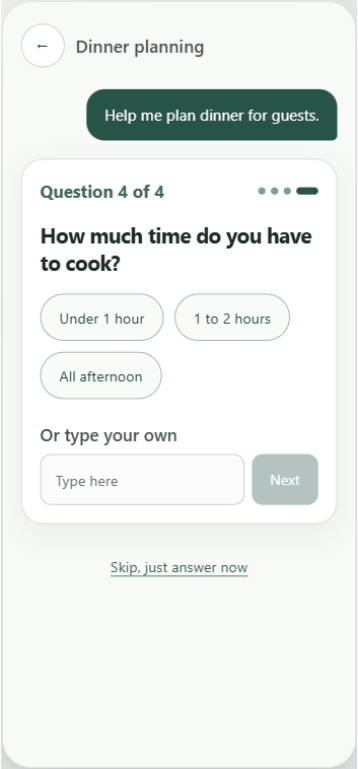
Welcome with Larger text switched on



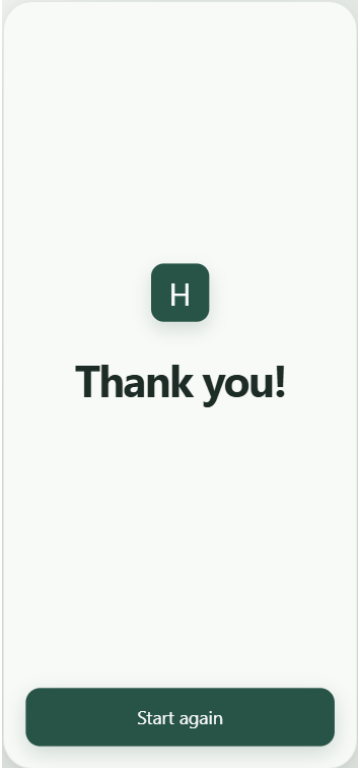
Setup with Larger text switched on



New culture question (question 3 of 4)



Time question, now question 4 of 4



New closing screen with a Start again button

Limitations

Only three people took part, all known to the researcher, and version 2 has not been retested, and their previous chatbot experience was not recorded, so it is not confirmed that they were novices. The prototype uses fixed text and no real model, so the study measures whether the interface is understood and liked, not whether it improves conversations. The researcher assisted one participant. The findings are exploratory and would need a larger and more varied sample, including all three settings, before any claim about satisfaction could be made.

References

[1] S.-H. Huang, Y.-F. Lin, Z. He, C.-Y. Huang, and T.-H. Huang, “How does conversation length impact user's satisfaction? A case study of length-controlled conversations with LLM-powered chatbots,” in Ext. Abstracts CHI Conf. Human Factors in Computing Systems (CHI EA '24), Honolulu, HI, USA, 2024. arXiv:2404.17025.

[2] H. Rahman and S. Desai, “Vibe check: Understanding the effects of LLM-based conversational agents' personality and alignment on user perceptions in goal-oriented tasks,” in Proc. CHI Conf. Human Factors in Computing Systems (CHI '26), Barcelona, Spain, 2026. arXiv:2509.09870.

[3] K. Bouyzourn and A. Birch, “What shapes user trust in ChatGPT? A mixed-methods study of user attributes, trust dimensions, task context, and societal perceptions among university students,” arXiv:2507.05046, 2025.

[4] N. Abarca, A. Carvallo, C. López Moncada, and F. Bravo-Marquez, “Explaining AI without code: A user study on explainable AI,” in LatinX in AI Workshop at NeurIPS 2025. arXiv:2602.11159.

1.5 Final research report (IEEE format)

The final report for paper 1 follows in IEEE two-column format.

Letting Novice Users Control Clarifying Questions in Chatbots: A Prototype and Small Usability Study

Wardah Ahmed
Applied Human-Computer Interaction (CS 3014), Class 5-B, Student ID 24K-1045
National University of Computer and Emerging Sciences (FAST NUCES), Karachi, Pakistan

***Abstract*—Chatbots that ask clarifying questions before they answer are now common. The evidence on how many questions users want comes mostly from young, educated people who already use chatbots, and the number was set by the researchers. This report asks how users with little chatbot experience respond when they control that number themselves. We built Helper, a clickable mobile prototype in which the user chooses no questions, three questions or a longer set, and can skip or change the choice. Three people aged 16, 23 and 47 tried it on a dinner-planning task. Two chose three questions and one chose none, and satisfaction ratings were 4, 3 and 1 out of 5. The main problems were not about the number of questions. Two participants mistook the answer for a summary, two did not notice the control for changing the setting, and one could not read the small text. We revised the prototype to address these problems. With three participants the findings are exploratory.**

***Index Terms*—conversational agents, clarifying questions, user control, novice users, usability evaluation, prototype.**

I. INTRODUCTION

Conversational interfaces built on large language models have moved quickly from research prototypes to everyday tools. One design choice is how long a conversation runs before the system gives its answer. A chatbot can respond at once, or it can ask clarifying questions first, and the two feel different to the person using them.

Huang et al. [1] tested this with 24 participants and two GPT-4 chatbots on Slack. For questions that invite conversation, satisfaction rose from 3.64 with an immediate answer to 4.33 with seven follow-up questions ($p = 0.041$). The participants were young, well educated and, as university students, comfortable with chatbots, and the researchers fixed the number of questions in advance. A first-time user may not know why a chatbot is asking questions back, and may read a follow-up as the system failing to understand them.

This report asks one question: how do users with little chatbot experience respond to a chat interface in which they control the number of clarifying questions? We built a clickable prototype, tested it with three people, and revised it using what they told us.

II. RELATED WORK

Huang et al. [1] found that satisfaction rose with conversation length only for conversational questions, and only the seven-follow-up condition reached significance. Crowd workers rating the transcripts saw no consistent quality gain, and participants disagreed: some found longer exchanges engaging and others found them repetitive, depending on how urgent the question was. A result from 24 people supports a probable conclusion, not a guaranteed one.

Other work shows that user differences change how a chatbot is received. Rahman and Desai [2] had 150 participants use a travel-planning chatbot at low, medium or high personality intensity. The medium version was rated best on most measures. Bouyzourn and Birch [3] surveyed 115 students and found that frequent use of ChatGPT went with higher trust, while a better understanding of how language models work went with lower trust. Abarca et al. [4] compared 10 novices and 10 experts using an explainability module. Novices rated the explanations useful and trustworthy, and experts were more critical of how complete they were. Novices and experienced users therefore do not respond to the same interface in the same way. None of these studies tests clarifying questions with novices.

III. RESEARCH GAP AND QUESTION

Existing work on conversation length [1] tested experienced users and fixed the number of follow-up questions in advance. Related studies [2]-[4] show that user traits and experience change reactions to chatbots, but none examines how people with little chatbot experience respond to clarifying questions, or whether letting them control the number helps. Our research question is: how do users with little chatbot experience respond to a chat interface in which they control the number of clarifying questions?

IV. PROTOTYPE

A. Design

Helper is a clickable mobile prototype built in Figma Make. It has no real language model behind it: replies are fixed text, and buttons link one screen to the next. The demo task is planning a dinner for guests, which naturally invites questions. After a welcome screen, the user chooses how much Helper asks before answering: “Just answer” (no questions), “Ask me a few things” (three) or “Ask me everything you need” (the full set). Each question appears as a card with a counter, quick-reply chips, a text box and a “Skip, just answer now” link. The answer screen offers a “Change how much I ask you” control, and a feedback screen asks for a satisfaction rating from 1 to 5 and whether the number of questions was too few, about right or too many. All wording is plain English with no technical terms, and the setup choices mirror the 0, 3 and 7 follow-up conditions of Huang et al. [1].

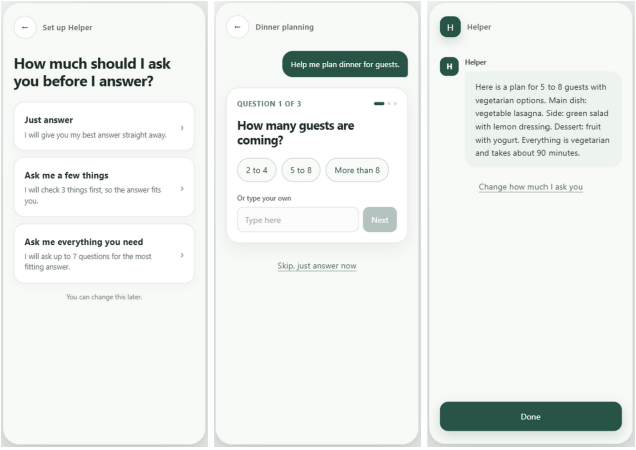


Fig. 1. Version 1 screens: setup, a clarifying question, and the answer.

B. Version 2

After the sessions we revised the prototype. Text size and contrast were increased on all screens, and a “Larger text” button was added to the welcome screen. The answer is now shown under a “Helper” label as a card headed “Your dinner plan”, the change control became a visible button, and a back arrow and a closing screen with a “Start again” button were added after the feedback screen. A question about food tradition was added, and the longest setting now asks up to four questions.

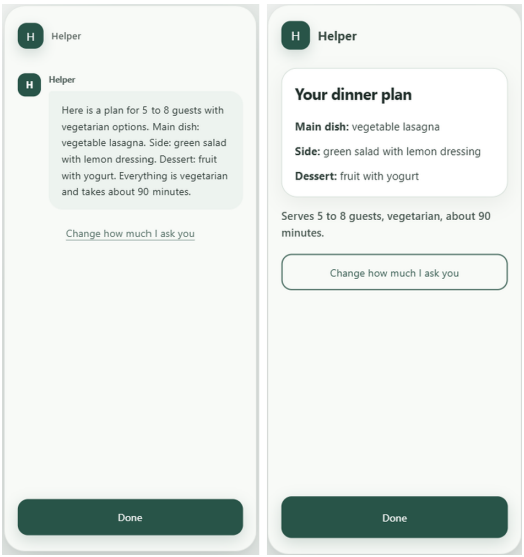


Fig. 2. The answer screen in version 1 (left) and version 2 (right).

V. EVALUATION

Three people took part: aged 16, 23 and 47. They were people the researcher knew, and their previous chatbot experience was not recorded. Each used the version 1 prototype on a phone, one at a time, with the researcher present. They were asked to imagine planning a dinner for guests, to use the app as they normally would and to think aloud. The researcher did not explain the settings and helped only when a participant was stuck. Afterwards, participants answered the two questions on the feedback screen and gave short spoken comments. The researcher recorded the option chosen, whether Skip was used and whether the change control was noticed.

VI. RESULTS

TABLE I. Observations from the three sessions. n/a: no questions were shown. Not rec.: not recorded.

ID	Age	Option	Skip	Change	Sat.	Questions
P1	16	Few (3)	No	No	4	Too few
P2	23	Few (3)	No	No	3	About right
P3	47	Just answer	n/a	Not rec.	1	About right

Two participants chose “Ask me a few things” and one chose “Just answer”. Nobody chose the longest setting, so it was not tested. Satisfaction was 4, 3 and 1. P1 found three questions too few and wanted more of them, in particular about culture and ethnicity. P2 first thought the questions were too few, then too many, then realised there were only three and said it did not matter. No one used Skip, which may mean it was not needed with only three questions.

Two problems appeared in the answer screen. P1 and P2 did not see the meal plan as a plan: P2 took it for a summary of her earlier choices, and P1 pressed Done without registering it. P2 suggested that the message should start below the “Helper” label so it is clear it comes from the app. P1 and P2 also did not notice the change control, and P1 pointed out there was no way back after Done.

P3 had difficulty reading the small on-screen text, and the researcher read the options aloud to her. Her rating of 1 reflects the difficulty of reading the screen. She skipped the questions, so it should not be read as a response to the question-length setting. P1 said she could use something like this.

VII. DISCUSSION

The three sessions give little evidence about the number of questions itself. Participants differed, with one wanting more and one finding the amount fine, which fits the idea of letting users decide, but a sample this small cannot show that control raises satisfaction. What the sessions did show is that interface details can dominate the experience. A first-time user read the answer as a summary of her own clicks, so the system's reply was not recognised as a reply. The control that would let people change their choice went unnoticed, and small text made the app hard to use for a participant who found it hard to see. For novice and older users, these basics may matter more than the question count.

P1’s request for questions about culture suggests that what is asked may matter as much as how many questions are asked. This comes from one participant and should be tested before it is treated as a finding. Design implications are to label system messages clearly, give a task result a distinct layout, make controls visible, keep text large and high in contrast, and give a way back after the final step.

VIII. LIMITATIONS AND FUTURE WORK

Only three people took part, all known to the researcher, and their chatbot experience was not recorded, so it is not confirmed that they were novices. The study had no group of experienced users, so the comparison implied by the gap is still open. The prototype uses fixed text and no real model, so it measures whether the interface is understood and liked, not whether it improves conversations. The researcher assisted one participant, and version 2 has not been retested. Future work should retest version 2 with a larger and more varied sample that includes experienced users and all three settings, connect the interface to a real model, and vary the number and content of questions.

IX. CONCLUSION

We built a prototype that lets users choose how many clarifying questions a chatbot asks and tested it with three people. The sessions showed that novice users may be held back first by the presentation of the answer, the visibility of controls and the legibility of text, and only after that by the number of questions. These observations are exploratory, but they give a concrete set of design changes and a basis for a larger study.

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[1] S.-H. Huang, Y.-F. Lin, Z. He, C.-Y. Huang, and T.-H. Huang, “How does conversation length impact user's satisfaction? A case study of length-controlled conversations with LLM-powered chatbots,” in *Ext. Abstr. CHI Conf. Human Factors in Computing Systems (CHI EA '24)*, Honolulu, HI, USA, 2024, arXiv:2404.17025.

[2] H. Rahman and S. Desai, “Vibe check: Understanding the effects of LLM-based conversational agents' personality and alignment on user perceptions in goal-oriented tasks,” in *Proc. CHI Conf. Human Factors in Computing Systems (CHI '26)*, Barcelona, Spain, 2026, arXiv:2509.09870.

[3] K. Bouyzourn and A. Birch, “What shapes user trust in ChatGPT? A mixed-methods study of user attributes, trust dimensions, task context, and societal perceptions among university students,” arXiv:2507.05046, 2025.

[4] N. Abarca, A. Carvallo, C. López Moncada, and F. Bravo-Marquez, “Explaining AI without code: A user study on explainable AI,” in *LatinX in AI Workshop at NeurIPS 2025*, arXiv:2602.11159.

Paper 2: DuetUI

Xu, Xiang, Song, Sun, and Tong. CHI '26, Barcelona, April 2026. HKUST (Guangzhou). arXiv:2509.13444.

2.1 Literature review

LLM agents are increasingly asked to carry out multi-step tasks for users, but they work best when the goal is clear from the start. When intent is vague or changes partway through, a single text prompt is a poor way to express it. Xu et al. [1] address this with a human-agent co-generation paradigm, implemented in a system called DuetUI.

In DuetUI the agent breaks a task down into an interface, and the user's direct manipulation of that interface steers what the agent generates next. The authors describe the agent and the user as collaboratively shaping both the task and the interface [1], so the goal itself can become clearer while the user works. The system rests on four features: a six-stage path (Define, Empathize, Plan, Explore, Refine, Duet), UI controls that let users direct the agent without typing prompts, a shared log of agent and user actions that gives the agent its context, and a link between the task breakdown and the interface description that keeps the two matched.

The design was shaped by an earlier study with 12 users, in which participants described the AI as a black box and doubted it could understand their instructions. The authors treat this as a gulf of execution [4]. Their second design goal responds with Shneiderman's direct manipulation [5], giving users a continuous representation of task objects and rapid, reversible actions [1]. The finished system was evaluated with a technical ablation study and a user study of 24 participants (12 female, aged 18 to 31) recruited through campus social media. The authors report gains in task efficiency and interface usability. Twenty participants said the system's lower barrier to entry removed the need for specialized knowledge, and eight praised its all-in-one workflow [1].

Trust was the most prominent theme in the user study. A majority of participants (16 of 24) called it a paramount concern, rooted in earlier experiences with LLM hallucinations and in moments during the study when simulated data looked like factual errors [1]. Participants also worried about the AI leaving out critical information, misreading important details, or using outdated data. They described trust as fragile. P14 said, “The incorrect information directly led to the collapse of my trust in this system... and that trust is very hard to win back” [1]. DuetUI already cites its data sources, and five participants noticed and appreciated this, which suggests that transparency is a direct route to trust. Participants proposed further mechanisms, including direct links to original sources, a preference for official data sources, a secondary verification agent, and peer review. The authors conclude that simulation alone did not resolve the trust gap, and suggest that future deployments add real-time source verification and iterative error repair [1].

The authors acknowledge other limitations. The participant pool was small and limited in age, demographics, and location. The system keeps a history of actions within a session but has no longer-term memory attached to the user, which the authors propose to add. Inclusivity also came up in the study itself. One participant (P4) cautioned that some groups, such as older adults, might still face learning difficulties, saying, “I'm worried my mom couldn't use this” [1].

The trust findings connect to the other studies in this set. Bouyzourn and Birch [2] found that trust in ChatGPT rose with frequent use and fell with a better understanding of how language models work, so the wariness seen in DuetUI fits a wider pattern. Abarca et al. [3] found novices and experts reacting differently to the same explanations, which shows that who the users are changes how a system lands.

2.2 Problem identification and research gap

Problem identification

Trust in DuetUI is fragile. Most participants arrived wary of hallucinations, and one incorrect output was enough for P14 to lose trust in the whole system. Citing sources helped, but it does not say what should happen after an error has already appeared. The authors point to real-time source verification and iterative error repair, and participants suggested versions of their own, such as links to original sources and a secondary verification agent. None of these were built or tested, so it is unknown whether they bring trust back once it has dropped.

Research gap

The authors found trust to be a main concern in [1] and observed that it persisted with simulated data. They propose source verification and error repair as future work. DuetUI already cites its data sources, but there is no evidence on whether verification and repair features restore a user's trust after the interface has shown an error. This project tests that. The study in [1] also had participants aged 18 to 31 recruited through campus social media, so the findings may not hold for older users, a limit that this project does not address directly.

Research question

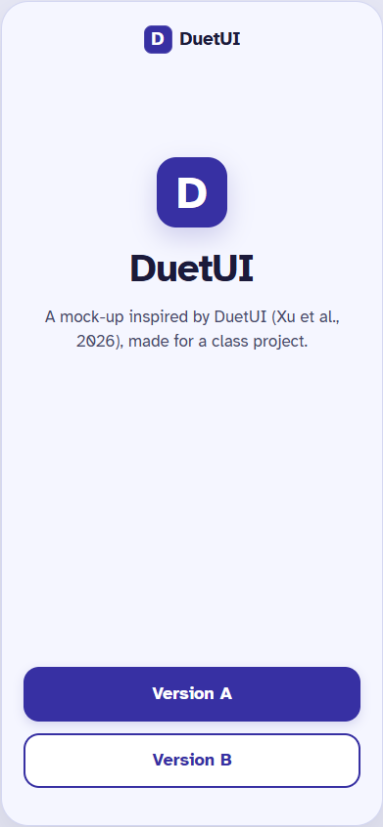
Does a co-generated interface that offers source links and a verify-and-repair step help users recover trust after the agent makes an error, compared with the same interface without them? The prototype in section 2.3 is built to answer this.

2.3 Prototype

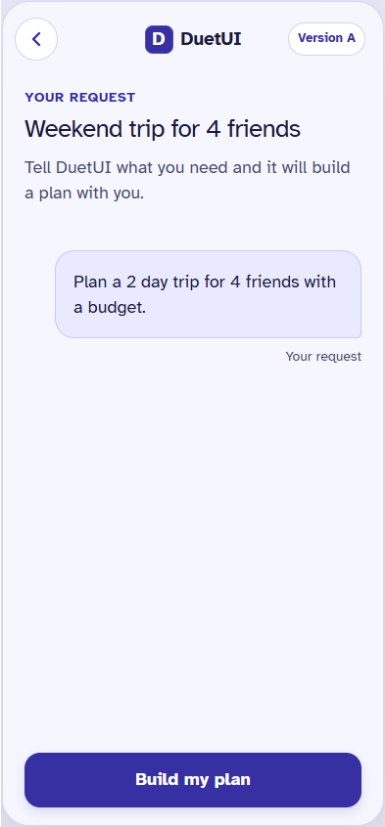
The prototype is a clickable mobile mock-up inspired by DuetUI [1], built in Figma Make. It is not the authors' system. It borrows one idea from DuetUI, an interface the agent builds together with the user and a visible history of what the agent did, and uses it to test the gap in section 2.2: whether source links and a verify-and-repair step help users recover trust after the agent makes an error. Nothing is connected to a real agent, and all content is fixed sample data that is labelled as such on screen. Prototype link: Figma Make prototype.

Both versions use the same task, planning a 2 day trip for four friends with a budget, and the same deliberate error: the four items add up to Rs 46,000, but the total reads Rs 41,000. In Version A (without verification), the user can only press “Looks good” or “Regenerate”, and Regenerate shows the corrected total with no explanation. In Version B (with verification), each item has a Source link that opens a short note, “Check sources” lists the sources and flags that the total does not match, and “Fix this” corrects it. An action history then records “DuetUI corrected the total from Rs 41,000 to Rs 46,000 (sum of items)”, in the spirit of the action history feature in DuetUI [1]. Each version ends with the same trust questions: a rating from 1 to 5 of trust before the user noticed anything wrong, a rating of trust now, and an open question about what would make them trust it more. Each participant sees only one version, so the ratings are compared between the two groups.

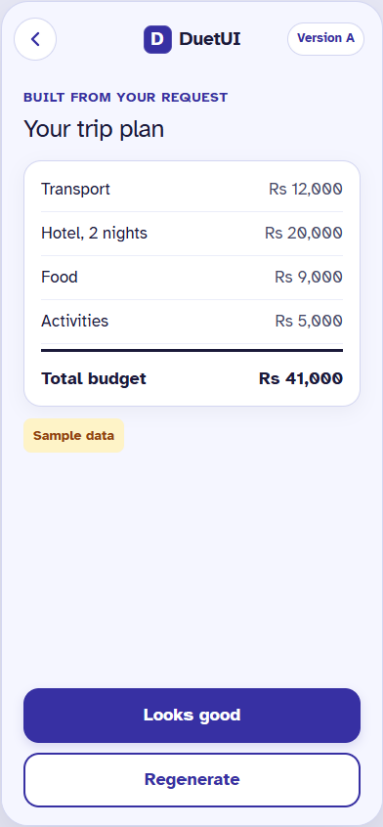
Design choices follow the earlier study in this report: text is large and high in contrast, because a participant there could not read small text, and the request is a fixed message bubble, not a text box, because the replies are fixed. A further screen in Version B, reached by pressing “Looks good” without checking, shows the same plan and leads to the same questions and is not pictured.



Start screen, where the researcher picks Version A or B



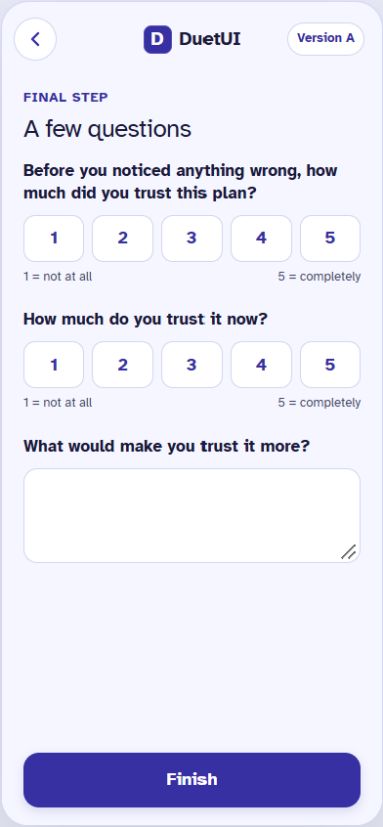
Version A: the task request



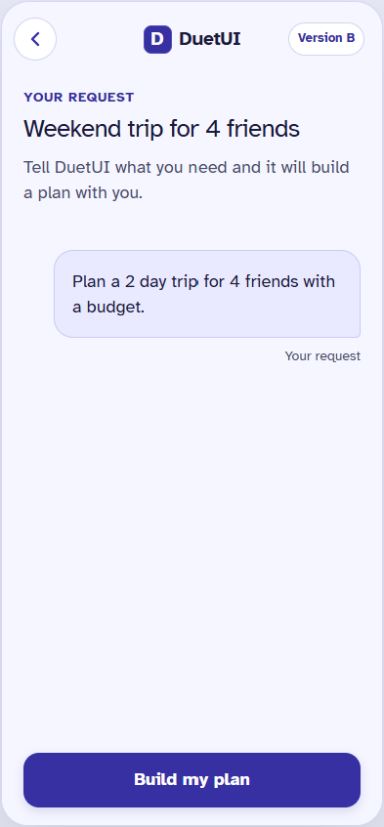
Version A: the plan, with the wrong total



Version A: after Regenerate, the total is corrected



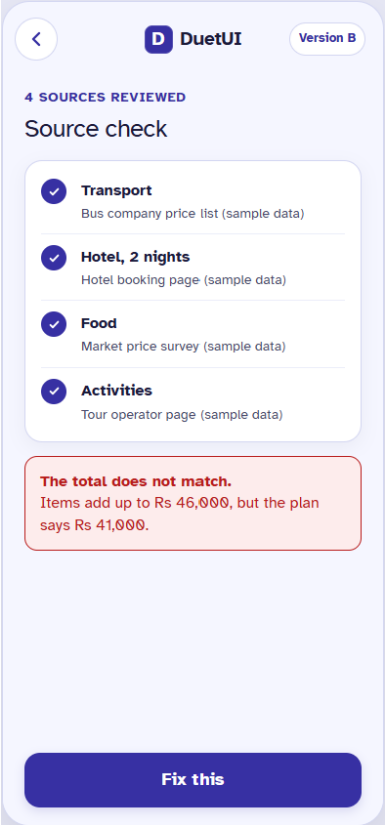
Version A: trust questions



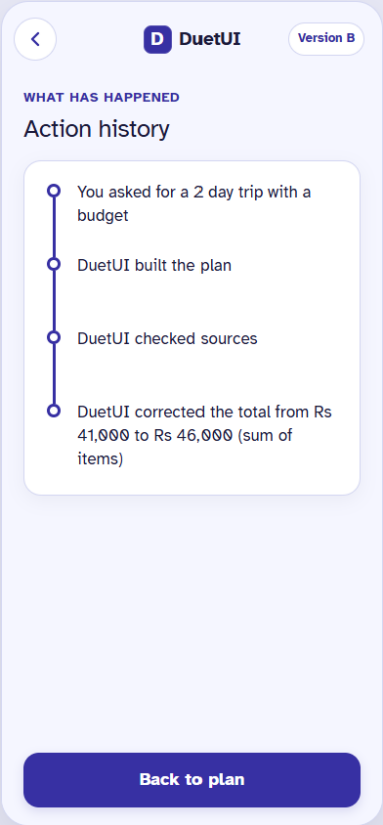
Version B: the task request



Version B: the plan with Source links and Check sources



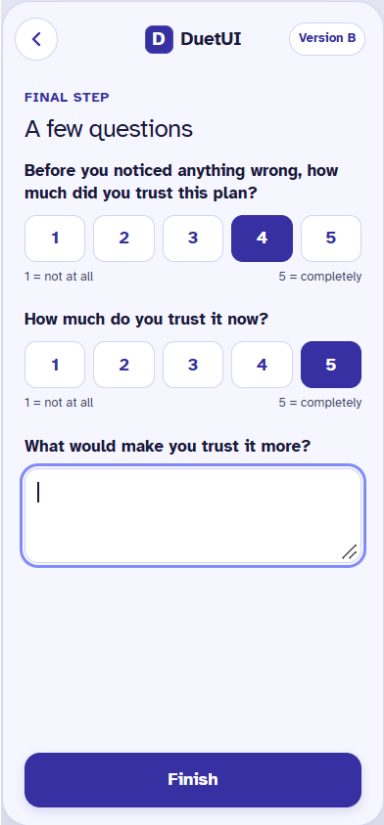
Version B: source check flags the mismatch



Version B: action history after the fix



Version B: the corrected plan



Version B: trust questions

2.4 User study and evaluation

The prototype was evaluated in a small, exploratory study with four participants. The aim was to see whether people noticed the planted error, how their trust in the plan changed, and whether the source and history features in Version B were used. Four people cannot support statistical claims, so the findings below are observations.

Procedure

Each participant used one version only, two on Version A and two on Version B, because the same error appears in both versions and a second run would give it away. The planned instruction was to look over the plan the way they would before booking a trip, without any hint that something was wrong. After the plan they answered the trust questions: a rating from 1 to 5 of trust before they noticed anything wrong, a rating of trust now, and an open question about what would make them trust the plan more. The researcher recorded which buttons each participant pressed and whether the participant noticed the wrong total. A line saying “you can use any button on the screen” was added during the study, so it may not have been given to every participant. The exact wording of the instructions was not recorded for each session.

Table 1. Observations from the four sessions. Ver.: version of the prototype. Before and Now: trust ratings from 1 to 5 before the error was noticed and after the plan.

ID	Age	Ver.	Noticed error	Pressed	Before	Now	Comment
A1	20	A	Yes	Regenerate	1	1	Needs a reasonable ground for price estimates, or a trusted brand behind them
A2	20	A	No	Looks good	2	2	Should specify details of the hotel or accommodation
B1	21	B	Not recorded	Looks good	3	3	Reviews from the public
B2	18	B	Not recorded	Check sources, Action history	3	4	Should show what the price includes, such as activities and food

Findings

Noticing the error. Only A1 is recorded as noticing the wrong total, and A2 did not. It was not recorded for B1 and B2. Because most participants were not told to look for a mistake, noticing it was left to them, and in Version B the mismatch is only flagged if the user presses “Check sources”. B1 pressed “Looks good” and so never saw that flag. The researcher's impression was that B1 did not understand what the task was about.

Trust in Version A. A1 noticed the error, pressed Regenerate and saw the corrected total with no explanation, and still rated trust 1 before and 1 now. A2 pressed “Looks good” without noticing the error and rated 2 and 2, so the low rating was not caused by the wrong total. For A1 the silent correction did not bring trust back, which is consistent with the gap in section 2.2, but it is one person.

Trust in Version B. B1 accepted the plan without checking and rated 3 and 3. B2 was the only participant to use the verification features. B2 pressed “Check sources”, opened the Action history, and rated trust 3 before and 4 now. This is the only increase in the study. The researcher noted that B2 did not care about the budget and was comfortable with prices changing, so the error may not have lowered B2's trust in the first place, and the one-point rise

cannot be treated as recovery.

Comments. All four participants said the plan needed a clearer basis. A1 asked for a reasonable ground for price estimates or a trusted brand, A2 for details of the hotel, B1 for reviews from the public, and B2 for what the price includes. None asked for a fix to the arithmetic. This suggests that these users were looking for what the plan rests on more than for a correction of one number.

Possible changes

The sessions point to several changes that were not built into the prototype. The mismatch could be flagged on the plan screen without the user having to press a button. Each price could say what it includes, and the plan could show public reviews or a known brand next to hotels and activities. The first two changes are small, so a second version of the prototype was not built.

Limitations

Four people aged 18 to 21 took part, two per version, so differences between the versions may reflect the people and not the design. Whether B1 and B2 noticed the error was not recorded, and the wording of the instructions was not recorded for each session. Only one participant used the verification features, so the study cannot say whether source checking and repair restore trust. The prototype uses fixed sample data. The findings are descriptive, and a larger and more varied sample, with each participant's use of the verification features recorded, would be needed for any claim about recovery.

References

[1] Y. Xu, S. Xiang, Y. Song, R. Sun, and X. Tong, “DuetUI: A bidirectional context loop for human-agent co-generation of task-oriented interfaces,” in Proc. CHI Conf. Human Factors in Computing Systems (CHI '26), Barcelona, Spain, 2026. arXiv:2509.13444.

[2] K. Bouyzourn and A. Birch, “What shapes user trust in ChatGPT? A mixed-methods study of user attributes, trust dimensions, task context, and societal perceptions among university students,” arXiv:2507.05046, 2025.

[3] N. Abarca, A. Carvallo, C. López Moncada, and F. Bravo-Marquez, “Explaining AI without code: A user study on explainable AI,” in LatinX in AI Workshop at NeurIPS 2025. arXiv:2602.11159.

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2.5 Final research report (IEEE format)

The final report for paper 2 follows in IEEE two-column format.

Source Checking and Trust in a Co-Generated Plan: A Prototype and Exploratory Study Inspired by DuetUI

Wardah Ahmed
Applied Human-Computer Interaction (CS 3014), Class 5-B, Student ID 24K-1045
National University of Computer and Emerging Sciences (FAST NUCES), Karachi, Pakistan

***Abstract*—Language model agents can now build task interfaces together with the user, but users' trust in them is fragile. The DuetUI study found trust to be the main concern and suggested source verification and error repair as future work, without testing them. We built a clickable mock-up inspired by DuetUI in two versions of one trip-planning task, each containing a planted arithmetic error. Version A allowed only acceptance or regeneration. Version B added source links, a source check that flags the mismatch, a fix button and an action history. Four participants aged 18 to 21 each used one version and rated their trust before and after. Only one of the four was recorded as noticing the error without help. The participant who regenerated in Version A rated trust 1 before and 1 after. The only participant who used the verification features rated trust 3 before and 4 after, but was relaxed about the budget. All four asked for a clearer basis for the plan. The findings are exploratory.**

***Index Terms*—human-agent collaboration, trust, error recovery, source verification, generated interfaces, prototype.**

I. INTRODUCTION

Language model agents are increasingly asked to carry out multi-step tasks, but a text prompt is a poor way to state a goal that is vague or that changes midway. Xu et al. [1] address this with DuetUI, in which the agent builds a task interface with the user and the user's direct manipulation of that interface steers what the agent generates next.

Trust was the most prominent theme in their user study. A majority of participants called it a paramount concern, and one participant said that trust lost after incorrect information is very hard to win back [1]. The authors suggest real-time source verification and iterative error repair for future deployments, but these were not built or tested. This report asks whether such features help users recover trust after the agent makes a mistake.

II. RELATED WORK

DuetUI rests on four ideas: a staged path from defining the task to refining the result, interface controls that let users direct the agent without writing prompts, a shared log of agent and user actions, and a link between the task breakdown and the interface description [1]. The system was evaluated in a technical ablation study and a user study of 24 participants aged 18 to 31, recruited through campus social media. Participants worried about hallucinated or outdated data. DuetUI already cites its data sources, and five participants appreciated this [1].

Other studies in this area point the same way. Bouyzourn and Birch [2] found that trust in ChatGPT was highest for tasks whose output is easy to verify and lowest for sources, and they call for explicit accuracy cues. Abarca et al. [3] found that novices and experts reacted differently to the same explanations. None of these tests what happens to trust after an error has appeared.

III. RESEARCH GAP AND QUESTION

The authors of [1] found trust to be a main concern and proposed source verification and error repair, but there is no evidence on whether such features restore a user's trust after the interface has shown an error. Our research question is: does a co-generated interface that offers source links and a verify-and-repair step help users recover trust after the agent makes an error, compared with the same interface without them?

IV. PROTOTYPE

The prototype is a clickable mobile mock-up built in Figma Make. It is inspired by DuetUI and is not the authors' system. Nothing is connected to a real agent, and all content is fixed sample data that is labelled as such on screen. Both versions use the same task, planning a two-day trip for four friends with a budget, and the same planted error: the four items add up to Rs 46,000, but the total reads Rs 41,000.

A. Version A: without verification

The user sees the plan and can press “Looks good” or “Regenerate”. Regenerate shows the corrected total with no explanation.

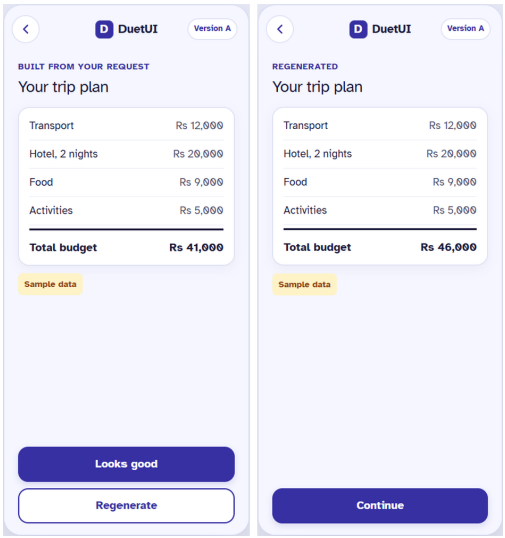


Fig. 1. Version A: the plan with the wrong total (left) and after Regenerate (right).

B. Version B: with verification

Each item has a Source link that opens a short note. “Check sources” lists the sources and flags that the total does not match, and “Fix this” corrects it. An action history then records that DuetUI corrected the total from Rs 41,000 to Rs 46,000, in the spirit of the action history in [1]. A user who presses “Looks good” skips all of this.

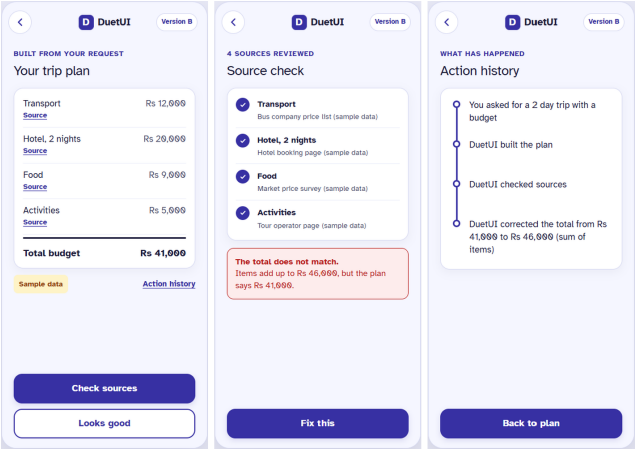


Fig. 2. Version B: the plan with Source links, the source check, and the action history.

Each version ends with the same questions: a rating from 1 to 5 of trust before the user noticed anything wrong, a rating of trust now, and an open question about what would make them trust the plan more. Text is large and high in contrast, and the request is a fixed message bubble because the replies are fixed.

V. EVALUATION

Four people aged 18 to 21 took part, two in each version. Each used one version only, because the same error appears in both and a second run would give it away. The planned instruction was to look over the plan as they would before booking a trip, with no hint that anything was wrong. The researcher recorded which buttons were pressed and whether the participant noticed the wrong total. A line saying participants could use any button was added during the study, so it may not have been given to everyone, and the exact wording of the instructions was not recorded for each session.

VI. RESULTS

TABLE I. Observations. A and B are the versions. Before and Now are trust ratings from 1 to 5. Not rec.: not recorded.

ID	Age	Noticed	Pressed	Before	Now
A1	20	Yes	Regenerate	1	1
A2	20	No	Looks good	2	2
B1	21	Not rec.	Looks good	3	3
B2	18	Not rec.	Check sources, history	3	4

Only A1 is recorded as noticing the wrong total, and A2 did not. A1 pressed Regenerate, saw the corrected total with no explanation, and still rated trust 1 before and 1 now. A2 pressed “Looks good” and rated 2 and 2, so the low rating was not caused by the wrong total. In Version B, B1 accepted the plan without checking and rated 3 and 3, and the researcher's impression was that B1 did not understand the task. B2 was the only participant to use the verification features, pressing “Check sources” and opening the action history, and rated trust 3 before and 4 now. The researcher noted that B2 did not care about the budget and was comfortable with prices changing.

All four participants said the plan needed a clearer basis. A1 asked for a reasonable ground for the prices or a trusted brand, A2 for details of the hotel, B1 for reviews from the public, and B2 for what the price includes. None asked for the arithmetic to be fixed.

VII. DISCUSSION

The silent correction in Version A did not restore trust for the one participant who noticed the error, which is consistent with the gap. The one increase in Version B came from a participant who was relaxed about the error, so it cannot be read as recovery. The sessions are more informative about two other points. First, finding the error depends on the user acting: in Version B the mismatch is flagged only after “Check sources” is pressed, and one participant who accepted the plan never saw it. Second, all four comments were about what the plan rests on, such as price grounds, hotel details, reviews and inclusions, and none were about the corrected number. These users appear to look for a basis for trust more than for a repair.

The changes suggested by the sessions are small: flagging the mismatch on the plan screen without a button press, and a line under each price saying what it includes. Because they are small, a second version of the prototype was not built. Showing public reviews or a known brand next to hotels and activities is a larger change and is left for future work.

VIII. LIMITATIONS AND FUTURE WORK

Only four people aged 18 to 21 took part, two per version, so differences between versions may reflect the people and not the design. Whether B1 and B2 noticed the error was not recorded, the instruction wording was not recorded for each session, and only one participant used the verification features, so the study cannot say whether source checking and repair restore trust. The prototype uses fixed sample data and no real agent. Future work should record noticing and button use for every participant, test the small changes above, include older users, and use a larger sample.

IX. CONCLUSION

We built two versions of a DuetUI-inspired planning mock-up to test whether source links and a verify-and-repair step help users recover trust after an error. With four participants, the study could not show recovery. It showed that users may not find the error unless the interface flags it, and that they want to know what a generated plan rests on. These observations are exploratory and give a direction for a larger study.

REFERENCES

[1] Y. Xu, S. Xiang, Y. Song, R. Sun, and X. Tong, “DuetUI: A bidirectional context loop for human-agent co-generation of task-oriented interfaces,” in *Proc. CHI Conf. Human Factors in Computing Systems (CHI '26)*, Barcelona, Spain, 2026, arXiv:2509.13444.

[2] K. Bouyzourn and A. Birch, “What shapes user trust in ChatGPT? A mixed-methods study of user attributes, trust dimensions, task context, and societal perceptions among university students,” arXiv:2507.05046, 2025.

[3] N. Abarca, A. Carvallo, C. López Moncada, and F. Bravo-Marquez, “Explaining AI without code: A user study on explainable AI,” in *LatinX in AI Workshop at NeurIPS 2025*, arXiv:2602.11159.

Paper 3: Vibe Check

Rahman and Desai. CHI '26, Barcelona, April 2026. Northeastern University. arXiv:2509.09870.

3.1 Literature review

Chatbots are often given a personality, a tone and style meant to make them feel more natural, but it is unclear how much personality helps. Rahman and Desai [1] study this with a travel-planning chatbot. Their design was a three-condition between-subjects experiment with 150 US adults recruited through Prolific, 50 per condition, all native English speakers aged 18 to 73, and most already familiar with chatbots (72 percent rated themselves very or completely familiar). Each participant planned a one-day trip to New York City in a text chat of up to 10 minutes with a GPT-4o chatbot. In the low, medium, and high conditions, all five Big Five traits were set to the same level through prompting [1].

The authors measured intelligence, anthropomorphism, likeability, enjoyment, intention to adopt, and trust. They also measured each user's own personality with a short questionnaire and calculated an alignment score for how close the user was to the chatbot. The personality condition significantly affected all six outcomes. The medium version beat the low version on every outcome and beat the high version on intelligence and likeability, which the authors describe as an inverted-U relationship between personality intensity and user experience. That reading rests on pairwise comparisons between the three conditions [1].

Alignment correlated modestly with the outcomes (Spearman correlations of about .23 to .30). After controlling for condition, most of the regression effects were not significant, and the authors describe them as directional trends that need larger samples. Participants fell into three clusters, and the well-aligned cluster, made up mostly of people in the medium condition, rated the chatbot higher than the globally misaligned cluster, made up mostly of people in the low condition. In comments, the medium chatbot was described as balanced and approachable, the low one as flat, and the high one as exhausting. The authors conclude that medium expression is the best default, name extraversion and emotional stability as the most consequential traits for alignment, and recommend adaptive calibration for the different compatibility profiles they found [1].

The authors state several limits. The sample was US adults who are native English speakers, which limits generalizability across cultures where personality expression and preferences may differ. Only one task, travel planning, was used, the 10-minute interaction cannot show how perceptions change over repeated use, and the vignette-based setup may not reflect real use where users have something at stake. The interface was text only, so the results may not carry over to voice. They also tested three uniform profiles out of 243 possible trait combinations, changing all five traits together, which they say “constrains theoretical precision” and means they “cannot isolate the independent contribution of any single trait.” They chose this deliberately, for ecological validity and as a first exploration, and note that single-trait manipulation has been shown to be feasible in earlier work. They add that prompting-based personality can vary together with behaviors such as sycophancy and praise, which makes it hard to attribute effects to personality alone [1].

Their proposed future work includes other task types, cultural and demographic differences, heterogeneous trait profiles, and longitudinal use. It also includes targeted manipulation of the traits behind composite behaviors such as politeness and sycophancy, to test how they affect

user perceptions and trust calibration [1].

The finding that more is not always better appears elsewhere in this set. Huang et al. [2] found that longer chatbot conversations raised satisfaction only for some kinds of questions, and Bouyzourn and Birch [3] found that human-likeness was not significantly related to overall trust in ChatGPT, with some interviewees finding human-like communication off-putting.

3.2 Problem identification and research gap

Problem identification

Because all five traits were changed together, the study can show that a medium amount of personality worked better, but not which trait did the work. The trait-level findings come from the distance between each user's trait score and the chatbot's, calculated after the fact, so they are correlations and cannot show that changing one trait caused a change in how users felt. The authors name extraversion as one of the two most consequential traits, and its distance from the user related to likeability, intelligence, and trust, yet it was never varied on its own.

Research gap

The authors say they “cannot isolate the independent contribution of any single trait” [1], and their future work points toward targeted trait manipulation. No study in this set isolates one trait while holding the others fixed. This project does that for extraversion, so that any difference in how users rate the chatbot can be traced to that one trait. It does not claim the idea is new, since the authors cite earlier single-trait work, only that it has not been tested on the question this paper raises.

Research question

Does changing only the extraversion of a goal-oriented chat assistant, with its other traits held constant, change how likeable and trustworthy users find it, and does the effect depend on the user's own extraversion? The prototype in section 3.3 is built to answer this.

3.3 Prototype

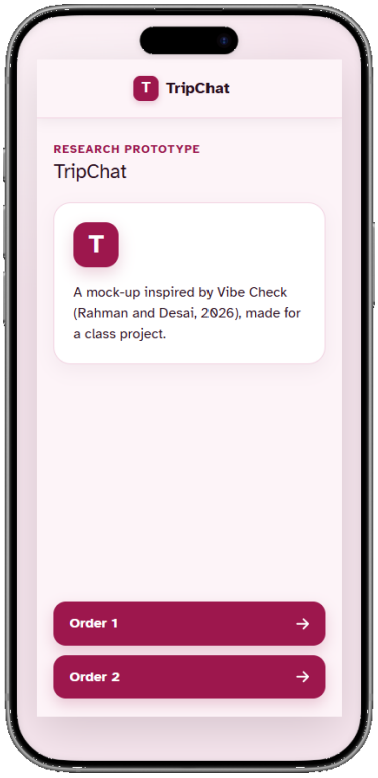
The prototype, TripChat, is a clickable mobile mock-up inspired by Vibe Check [1], built in Figma Make. It is not the authors' system and no real language model is connected. It uses one idea from the paper, giving a goal-oriented assistant a personality, and tests the gap in section 3.2 by changing only one trait. The task is the same as in the paper's setting, planning a weekend trip, and every user sees the same task in two short scripted chats.

The two assistants give the same information in the same order: a two day plan for Lahore, a dinner suggestion, and a closing reply. Only the manner of speaking differs. One chat is calm and reserved (extraversion low): short, neutral sentences with no exclamation marks. The other is lively and outgoing (extraversion high): enthusiastic openings such as “Great choice!”, exclamation marks, and a friendlier tone. The wording of both scripts is my own, written to represent low and high extraversion, and is not text taken from the paper. All other traits are kept the same by keeping the content, length and structure identical. A start screen offers Order 1 (calm first) and Order 2 (lively first), so the order can be alternated between participants to reduce order effects.

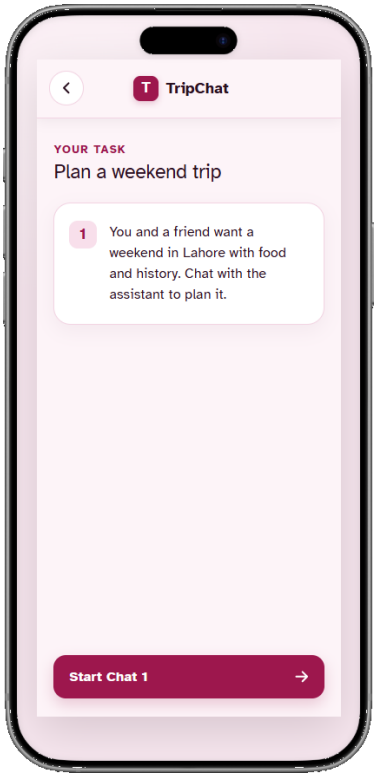
After each chat the user rates the assistant from 1 to 5 on how likeable it is and how much they trust it. After both chats they say which assistant they preferred, or that they have no

preference, and answer two short items about themselves (“I am outgoing and enthusiastic” and “I am reserved and quiet”) so that the effect can be compared with the user's own extraversion. Each participant therefore sees both versions, and the ratings are compared within each person.

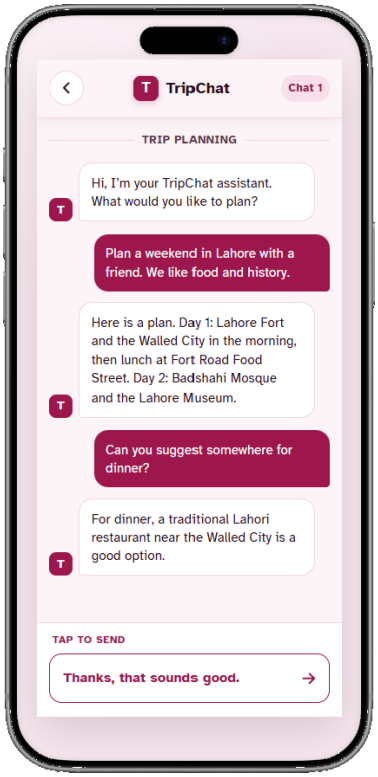
Prototype link: Figma Make prototype.



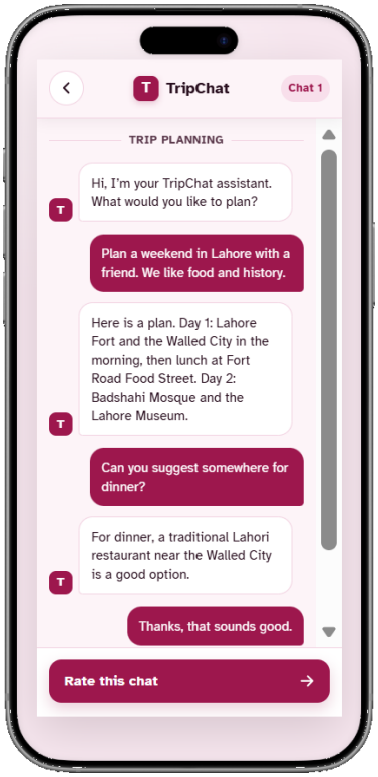
Start screen with the two orders



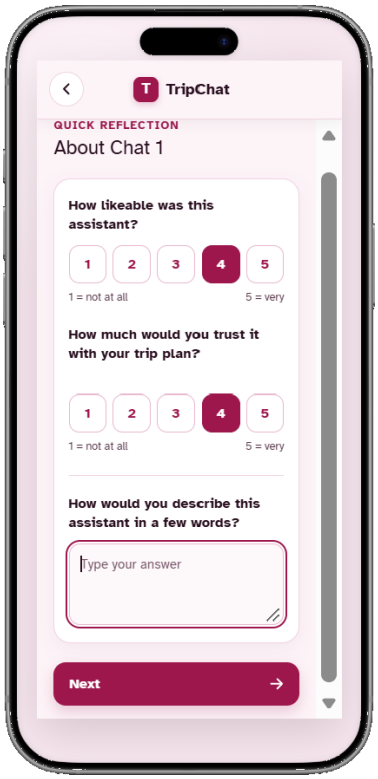
Task: plan a weekend trip



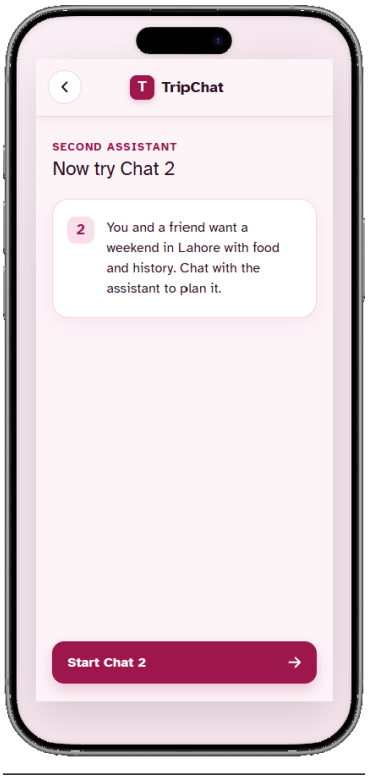
Chat 1 (calm), in progress



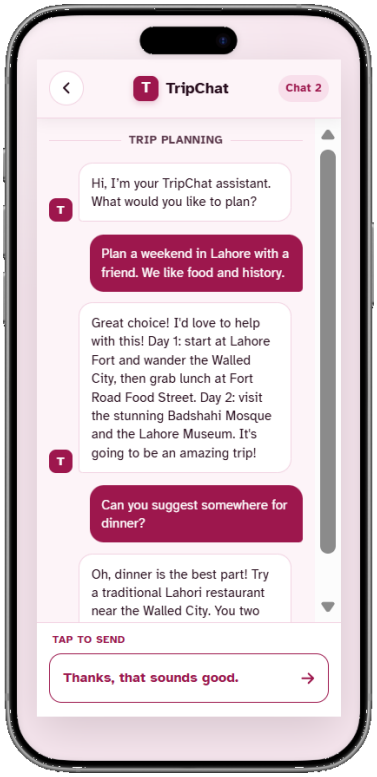
Chat 1, after the last message



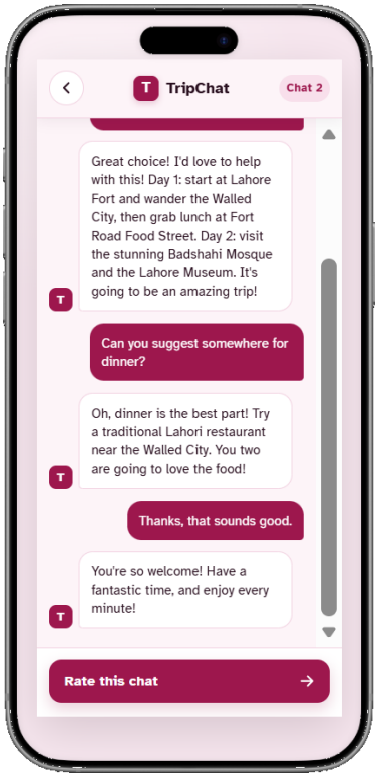
Ratings for Chat 1



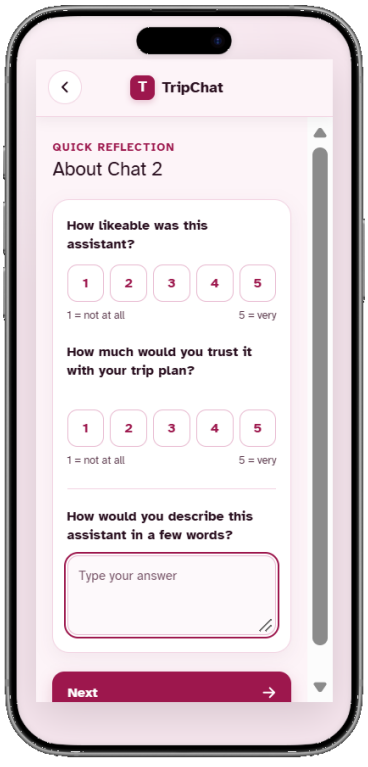
Prompt to start Chat 2



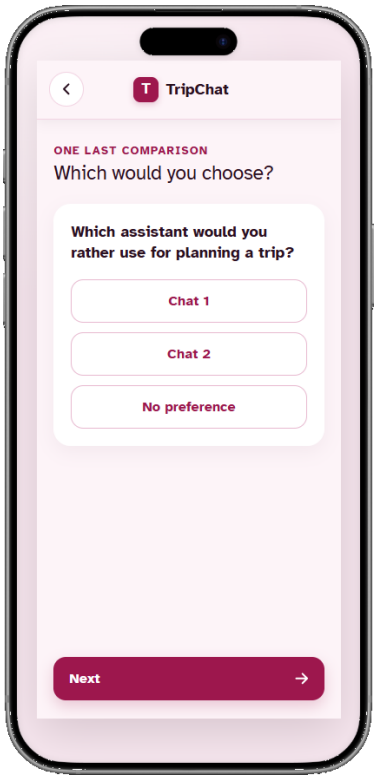
Chat 2 (lively), in progress



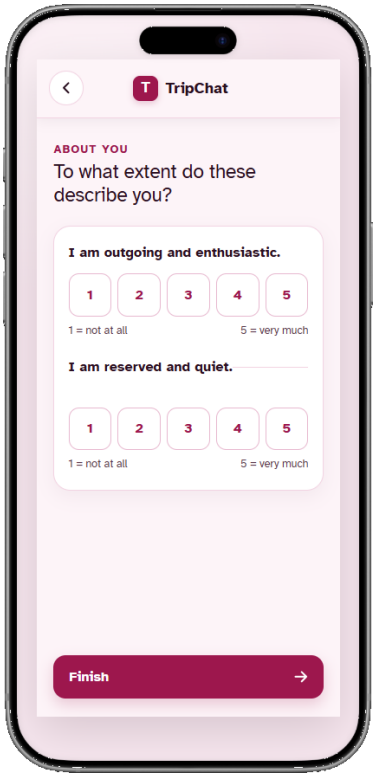
Chat 2, after the last message



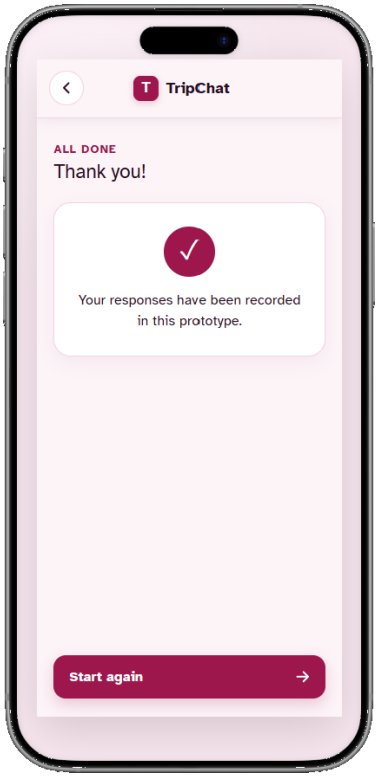
Ratings for Chat 2



Preference between the chats



Two questions about the user



Thank you screen

In the screenshots above, the Chat 1 rating screen shows an example answer already selected and the Chat 2 rating screen is empty. The screenshots show Order 1 only. In Order 2 the same two chats appear in the opposite order.

3.4 User study and evaluation

TripChat was tried by three participants in a small, exploratory study. The aim was to see whether people rated the lively assistant differently from the calm one on likeability and trust, which one they preferred, and whether this seemed to relate to how outgoing they said they were. Three people cannot support statistical claims, so the results below are observations.

Procedure

Each participant went through both chats, so every person rated both the calm and the lively assistant. Order 1 (calm first) was used for the first two participants and Order 2 (lively first) for the third. After each chat the participant rated likeability and trust from 1 to 5 and could write a few words describing the assistant. At the end they chose the assistant they preferred, or no preference, and answered the two extraversion items. Participants were not told which assistant was calm and which was lively. The only instruction was to click through the prototype, which takes about five minutes, and to give their responses.

Table 1. Results from the three sessions. L: likeable, T: trust (ratings from 1 to 5). Outgoing and Reserved: self-ratings from 1 to 5 on "I am outgoing and enthusiastic" and "I am reserved and quiet".

ID	Age	Order	Calm (L / T)	Lively (L / T)	Preferred	Outgoing / Reserved
P1	18	1	4 / 3	4 / 2	Calm	4 / 2
P2	20	1	2 / 2	3 / 3	Lively	4 / 3
P3	23	2	3 / 3	5 / 4	Lively	4 / 3

Findings

Ratings. Averaged over the three participants, the lively assistant was rated higher on likeability (4.0 against 3.0) and slightly higher on trust (3.0 against 2.7). The pattern is not the same for everyone. P2 and P3 rated the lively assistant higher on both questions. P1 gave both the same likeability rating and trusted the calm assistant more.

Preference. Two of three participants preferred the lively assistant and one preferred the calm one. Nobody chose “no preference”.

Order. The first chat was rated higher than the second by P1 and P3 and lower by P2, so there is no consistent order effect in these data, although three people are too few to rule one out.

Own extraversion. All three participants rated themselves 4 on “outgoing and enthusiastic” and between 2 and 3 on “reserved and quiet”, so nobody in the sample was clearly reserved. P1 and P2 gave almost the same self-ratings and still chose different assistants. The second half of the research question, whether the effect depends on the user's own extraversion, cannot be answered with this sample.

Comments. P3 left both comment boxes empty. P1 said the calm assistant did not ask what was feasible, such as where the hotel was and what the budget was, and that the lively assistant was full of energy when the focus should have been on the plan, which should be in depth and not only suggest places. P2 said both assistants were poor, with the lively one better. In P2's view both told the user what to do and did not offer choices, the calm one read like pre-written answers behind buttons, and the lively one used generic phrases about itself, such as saying it loves food, that do not help the user. P2 described what would work better: acknowledge what the user said they want, offer several options, and give a reason for each so that the user can change them. The three comments agree on one point: the plan itself and its fit to the user mattered more to these participants than the tone.

Possible changes

The comments suggest a second version in which both assistants ask for the user's hotel and budget, offer a few options with a reason for each, and let the user adjust them, with extraversion varied only in the wording around that content. This was not built, so the study here tests tone alone.

Limitations

Three people aged 18 to 23 took part, so any difference between the assistants may reflect the individuals and not the tone. The orders were not balanced (two participants had Order 1 and one had Order 2). All participants described themselves as fairly outgoing, and the self-ratings use two items and not a full personality scale. Both scripts were written for this project, so they show one version of low and high extraversion and not a validated one. Because the content was held identical, both assistants shared the same weaknesses, which participants pointed out and which may have limited how much the tone mattered. The instructions were minimal, and the results are descriptive only.

References

[1] H. Rahman and S. Desai, “Vibe check: Understanding the effects of LLM-based conversational agents' personality and alignment on user perceptions in goal-oriented tasks,” in Proc. CHI Conf. Human Factors in Computing Systems (CHI '26), Barcelona, Spain, 2026. arXiv:2509.09870.

[2] S.-H. Huang, Y.-F. Lin, Z. He, C.-Y. Huang, and T.-H. Huang, “How does conversation length impact user's satisfaction? A case study of length-controlled conversations with LLM-powered chatbots,” in Ext. Abstracts CHI Conf. Human Factors in Computing Systems (CHI EA '24), Honolulu, HI, USA, 2024. arXiv:2404.17025.

[3] K. Bouyzourn and A. Birch, “What shapes user trust in ChatGPT? A mixed-methods study of user attributes, trust dimensions, task context, and societal perceptions among university students,” arXiv:2507.05046, 2025.

3.5 Final research report (IEEE format)

The final report for paper 3 follows in IEEE two-column format.

Changing Only the Extraversion of a Trip-Planning Chat Assistant: A Prototype and Exploratory Study Inspired by Vibe Check

Wardah Ahmed
Applied Human-Computer Interaction (CS 3014), Class 5-B, Student ID 24K-1045
National University of Computer and Emerging Sciences (FAST NUCES), Karachi, Pakistan

***Abstract*—Giving a chat assistant a personality is common, but it is hard to tell which trait matters. Rahman and Desai changed all five Big Five traits together and state that they cannot isolate the contribution of any single trait. We built a clickable mock-up, TripChat, in which two scripted trip-planning assistants give identical content and differ only in extraversion: one calm and reserved, one lively and outgoing. Three participants aged 18 to 23 used both assistants, with the order varied, rated likeability and trust from 1 to 5, chose a preferred assistant, and rated their own extraversion. The lively assistant was rated higher on likeability (mean 4.0 against 3.0) and slightly higher on trust (3.0 against 2.7), and two of three participants preferred it. Comments from two participants said that neither assistant offered options with reasons, and that the plan mattered more than the tone. All participants described themselves as fairly outgoing, so the study cannot say whether the effect depends on the user. The findings are exploratory.**

***Index Terms*—conversational agents, personality, extraversion, likeability, trust, prototype.**

I. INTRODUCTION

Chat assistants are often given a personality, a tone and style meant to make them feel natural. Rahman and Desai [1] tested this with a travel-planning chatbot and found that a medium level of personality was rated best, but they changed all five Big Five traits together. They say this “constrains theoretical precision” and that they cannot isolate the independent contribution of any single trait [1]. This report asks what happens when only one trait is changed.

II. RELATED WORK

Rahman and Desai [1] ran a between-subjects experiment with 150 US adults. Each planned a one-day trip to New York City with a GPT-4o chatbot set to a low, medium, or high level of all five traits. The personality condition significantly affected intelligence, anthropomorphism, likeability, enjoyment, intention to adopt, and trust, and the medium version beat the low version on every outcome. The authors also measured the distance between each user's personality and the chatbot's. This alignment correlated modestly with the outcomes, and they name extraversion and emotional stability as the most consequential traits. Because the distances were calculated afterwards, these trait results are correlations and not effects of changing a trait.

The finding that more is not always better appears in other work. Huang et al. [2] found that longer conversations raised satisfaction only for some kinds of questions, and Bouyzourn and Birch [3] found that human-likeness was not significantly related to overall trust in ChatGPT, with some interviewees finding human-like communication off-putting.

III. RESEARCH GAP AND QUESTION

The authors of [1] call for targeted manipulation of single traits, and note that earlier work has shown it to be feasible. We do not claim the idea is new. We test it on the question their results raise: does changing only the extraversion of a goal-oriented chat assistant, with its other traits held constant, change how likeable and trustworthy users find it, and does the effect depend on the user's own extraversion?

IV. PROTOTYPE

TripChat is a clickable mobile mock-up built in Figma Make. It is inspired by Vibe Check and is not the authors' system. No language model is connected, and both assistants are fixed scripts. The task is planning a weekend trip, and each user sees it in two short chats.

The assistants give the same information in the same order: a two-day plan for Lahore, a dinner suggestion, and a closing reply. They differ only in manner. The calm assistant (low extraversion) uses short neutral sentences without exclamation marks. The lively assistant (high extraversion) opens with enthusiasm, uses exclamation marks, and is friendlier. The wording of both scripts was written for this project and is not validated text from [1]. Content, length and structure are identical, so that other traits stay constant. The start screen offers Order 1 (calm first) and Order 2 (lively first) so that order can be alternated.

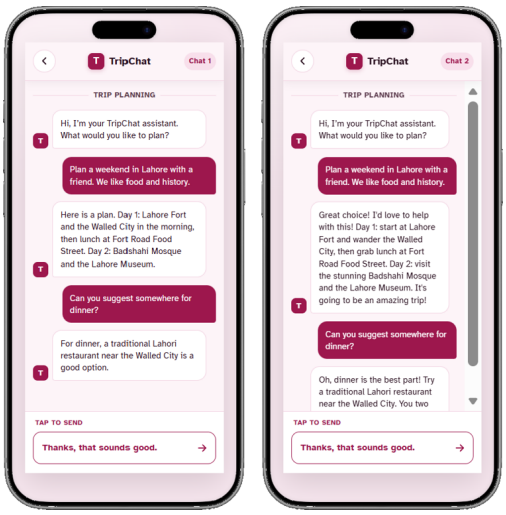


Fig. 1. The calm assistant (left) and the lively assistant (right) in Order 1.

After each chat the user rates likeability and trust from 1 to 5 and may describe the assistant in a few words. After both chats they choose the assistant they prefer, or no preference, and rate two statements about themselves: “I am outgoing and enthusiastic” and “I am reserved and quiet”.

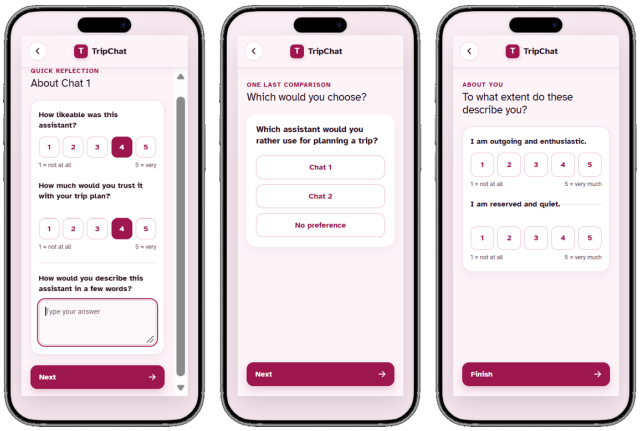


Fig. 2. The rating screen, the preference screen, and the self-rating screen.

V. EVALUATION

Three people aged 18 to 23 took part. Each used both assistants, so ratings are compared within each person. Two participants had Order 1 and one had Order 2. Participants were not told which assistant was calm and which was lively. The only instruction was to click through the prototype, which takes about five minutes, and to give their responses.

VI. RESULTS

TABLE I. Results. L: likeable, T: trust (1 to 5). Pref.: preferred assistant. Out./Res.: self-ratings of outgoing and reserved (1 to 5).

ID	Order	Calm L/T	Lively L/T	Pref.	Out./Res.
P1	1	4 / 3	4 / 2	Calm	4 / 2
P2	1	2 / 2	3 / 3	Lively	4 / 3
P3	2	3 / 3	5 / 4	Lively	4 / 3

The lively assistant was rated higher on likeability (mean 4.0 against 3.0) and slightly higher on trust (3.0 against 2.7). P2 and P3 rated it higher on both questions. P1 gave both the same likeability rating and trusted the calm assistant more. Two of three participants preferred the lively assistant, and nobody chose no preference. The first chat was rated higher than the second by P1 and P3 and lower by P2, so no consistent order effect appears, although three people cannot rule one out. All three rated themselves 4 on outgoing and 2 or 3 on reserved. P1 and P2 gave almost identical self-ratings and chose different assistants.

P3 left the comment boxes empty. P1 said the calm assistant did not ask what was feasible, such as the hotel and budget, and that the lively one was full of energy when the focus should be on an in-depth plan. P2 said both assistants were poor and the lively one was better: both told the user what to do instead of offering choices, the calm one read like pre-written answers behind buttons, and the lively one used generic phrases about itself, such as saying it loves food. P2 suggested that an assistant should acknowledge what the user wants, offer several options, and give a reason for each so the user can change them.

VII. DISCUSSION

With three participants, the study can only say that the lively assistant tended to be rated higher and that the result was not the same for everyone. The second half of the research question, whether the effect depends on the user's own extraversion, cannot be answered, because nobody in the sample was clearly reserved and the two participants with the same self-ratings preferred different assistants. The comments matter more than the numbers here. Both participants who commented said the plan and its fit to the user counted for more than the tone, and one described what a better assistant would do. This agrees with the authors' own caution that personality may vary together with behaviors such as sycophancy and praise [1]. Since the content was held identical, both assistants shared the same weakness, which may have limited how much tone could matter.

A second version would have both assistants ask for the user's hotel and budget, offer a few options with a reason for each, and let the user adjust them, with extraversion varied only in the wording around that content. This was not built, so the study tests tone alone.

VIII. LIMITATIONS AND FUTURE WORK

Three participants, all fairly outgoing and aged 18 to 23, took part, so differences may reflect the individuals and not the tone. The orders were not balanced. Extraversion was measured with two items, not a full scale. The two scripts are one version of low and high extraversion written for this project, not a validated one, and the instructions were minimal. No real language model was used. Future work should use a balanced order, recruit reserved participants, use a validated scale, and compare assistants that offer options and reasons.

IX. CONCLUSION

We built a mock-up that varies only extraversion in a trip-planning assistant. In a three-person exploratory study the lively assistant tended to be rated higher and preferred more often, but one participant reversed this, no reserved users took part, and participants said the content of the plan mattered more than the tone. A larger study with reserved participants and richer content is needed.

REFERENCES

[1] H. Rahman and S. Desai, “Vibe check: Understanding the effects of LLM-based conversational agents' personality and alignment on user perceptions in goal-oriented tasks,” in *Proc. CHI Conf. Human Factors in Computing Systems (CHI '26)*, Barcelona, Spain, 2026, arXiv:2509.09870.

[2] S.-H. Huang, Y.-F. Lin, Z. He, C.-Y. Huang, and T.-H. Huang, “How does conversation length impact user's satisfaction? A case study of length-controlled conversations with LLM-powered chatbots,” in *Ext. Abstracts CHI Conf. Human Factors in Computing Systems (CHI EA '24)*, Honolulu, HI, USA, 2024, arXiv:2404.17025.

[3] K. Bouyzourn and A. Birch, “What shapes user trust in ChatGPT? A mixed-methods study of user attributes, trust dimensions, task context, and societal perceptions among university students,” arXiv:2507.05046, 2025.

Paper 4: What Shapes User Trust in ChatGPT?

Bouyzourn and Birch. University of Edinburgh and KU Leuven. arXiv:2507.05046.

4.1 Literature review

Trust shapes whether people rely on tools like ChatGPT, but what drives it is less clear. Bouyzourn and Birch [1] study this with a mixed-methods design: a survey of 115 University of Edinburgh students (45 undergraduates, 36 taught postgraduates, and 34 research postgraduates, aged 18 to 35, 77 of them in computer science) recruited through course announcements and social media, plus four semi-structured interviews. The survey covered seven trust dimensions (expertise, predictability, transparency, human-likeness, ease of use, risk, and reputation), trust for 11 task types, and views on AI's societal impact [1].

Frequent users, who use it more than once a week, trusted ChatGPT more than infrequent users (mean 3.43 against 2.85, $p = .001$). Participants who said they understood how language models work trusted it less than those who did not or were unsure (mean 3.06 against 3.35, $p = .049$). Field of study and degree level made no significant difference. Of the seven dimensions, perceived expertise was the strongest predictor of overall trust ($B = 0.34$, $p < .001$), and human-likeness neither correlated with nor predicted trust. Trust varied by task. It was highest for summarising (3.05), explaining (3.02), and coding (3.01), and lowest for sources (2.19) and entertainment (2.22). Participants who saw AI's societal impact as positive trusted ChatGPT more than those who saw both benefits and harms ($p = .015$) [1].

The interviews added reasons. Participants described trust dropping after they found references that did not exist, and growing as they verified outputs. Some found human-like communication off-putting. The authors report that users with technical knowledge were more critical, while less experienced users tended to trust fluent outputs at first, and that both groups adjusted their trust through experience [1].

The authors also note a paradox. Although trust in reference generation was among the lowest, overall trust rose with confidence in generating references, despite well-documented error rates. They read this as automation bias, with persuasive fluency mistaken for factual reliability, and conclude that trust is calibrated by task verifiability, perceived objectivity, and user expertise. They point to the need for explicit accuracy cues and targeted user education, recommend that designers enable users to understand and assess AI outputs and support verification and explainability for high-stakes tasks, and recommend integrating AI literacy into university curricula [1].

The paper has no dedicated limitations section. From the design, the sample came from one university and was mostly computer science students, and the interview group was four people. The survey was cross-sectional and self-reported, so it cannot show how trust changes through use. These are our observations, not the authors' statements. The authors do say that future work should examine whether trust itself motivates continued use [1].

The results connect to other papers in this set. Abarca et al. [3] found novices trusting an explanation module more readily than experts, which parallels users who understand language models trusting ChatGPT less. Xu et al. [2] reported a participant whose trust in DuetUI collapsed after incorrect information, as an interviewee here lost trust after finding fabricated references.

4.2 Problem identification and research gap

Problem identification

Trust in ChatGPT is not uniform. On average it is high for summarising and low for finding sources, yet people who trust ChatGPT more overall also trust it more for generating references, which the authors read as automation bias. A chat interface presents every answer in the same way whatever the task, so users have to judge reliability for themselves, and the interviews suggest they do it by verifying outputs or by losing trust after being let down. The authors call for explicit accuracy cues, but the study tests no interface.

Research gap

The authors conclude that trust is calibrated by task verifiability and user expertise, and call for explicit accuracy cues and targeted user education [1]. They do not test whether an interface can help users match their trust to how reliable the tool is for each task. Their cross-sectional, self-reported design also cannot show whether such cues change trust. This project tests task-specific reliability cues. It differs from the DuetUI work in section 2, which concerns recovering trust after an error, by looking at setting trust appropriately before answers are used.

Research question

Does a chat interface that shows task-specific reliability cues help users bring their trust in line with how reliable the tool is for each task, compared with the same interface without the cues? The prototype in section 4.3 is built to answer this.

4.3 Prototype

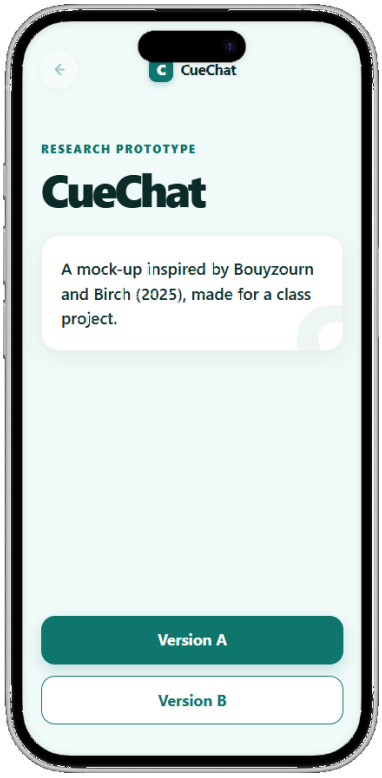
The prototype, CueChat, is a clickable mobile mock-up inspired by Bouyzourn and Birch [1], built in Figma Make. It is not a real assistant and nothing is connected to a language model. It tests the gap in section 4.2 by showing, for each answer, how reliable this kind of answer usually is, and then asking how much the user would rely on it. All content is fixed sample content, which is stated at the bottom of the chat screens. Prototype link: Figma Make prototype.

Every participant sees the same three tasks in the same order. The first asks the assistant to summarise a short paragraph about Mohenjo-daro, and the summary is accurate. The second asks it to explain overfitting in machine learning, and the explanation is correct. The third asks for three academic references on sleep and exam performance, and the three references were invented for this study. This matches the tasks in [1], where trust was highest for summarising and explaining and lowest for sources. After each answer the user rates how much they would rely on it, from 1 to 5. At the end they tick which answers they would double-check and describe what would help them decide how much to trust an answer like these. A debrief screen tells the participant that the references were made up.

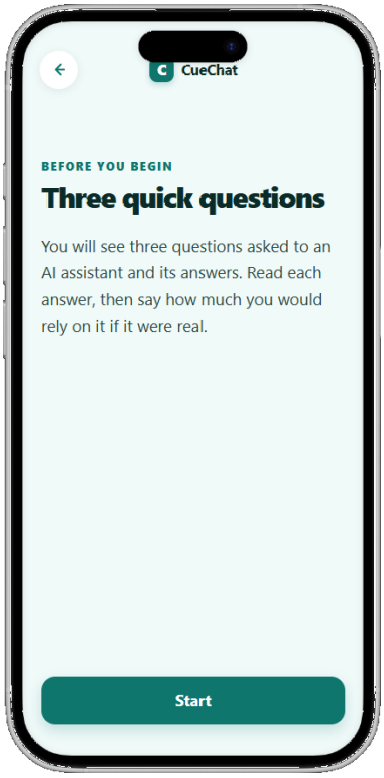
The prototype has two versions, and each participant sees only one. Version A shows the answer and the rating question. Version B adds a “Reliability for this task” card under each answer: a High badge for the summary, Medium for the explanation, and Low for the references, each with a one-line reason and a “How to check” tip. The three levels are our own judgement, based on the ordering of task trust reported in [1], and not a measured accuracy of any real assistant. If users in Version B rate the references lower than the summary by a wider

margin than users in Version A, that would suggest that the cues help trust follow reliability.

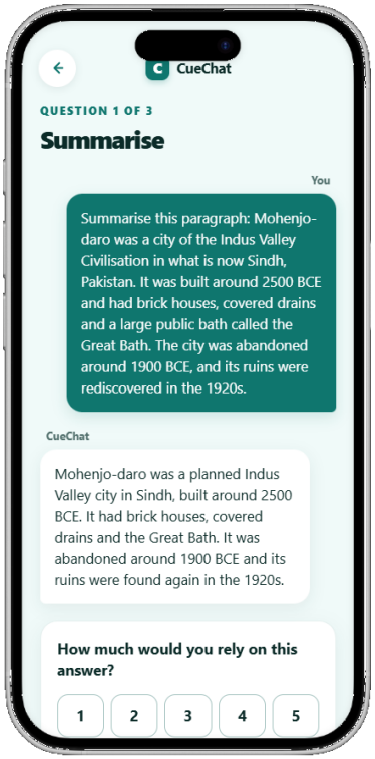
Version A (without reliability cues)



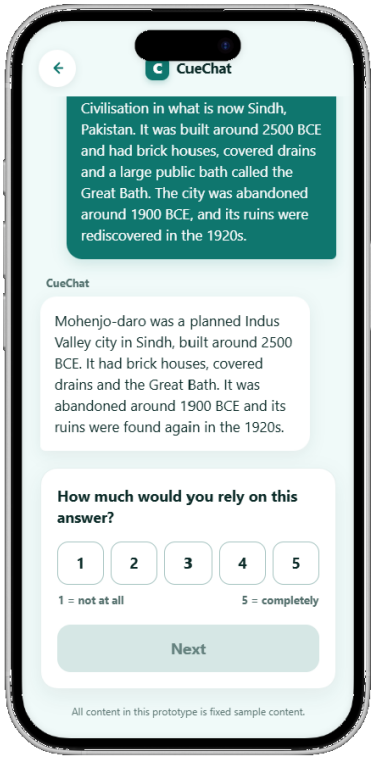
Start screen with the two versions



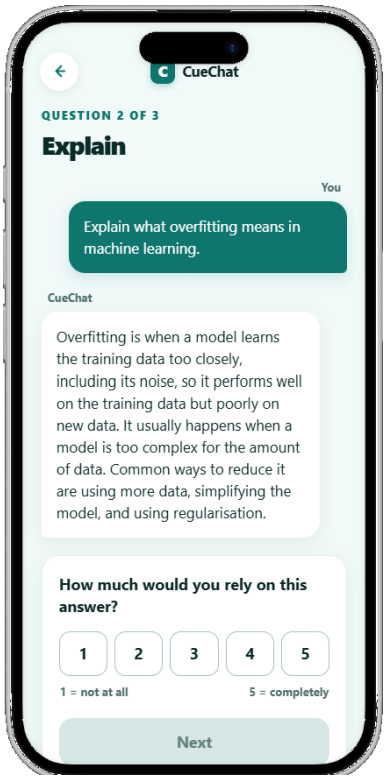
Introduction



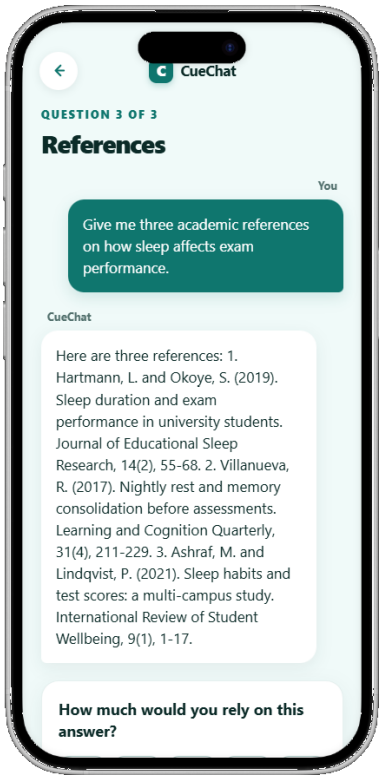
Question 1, summary: the answer



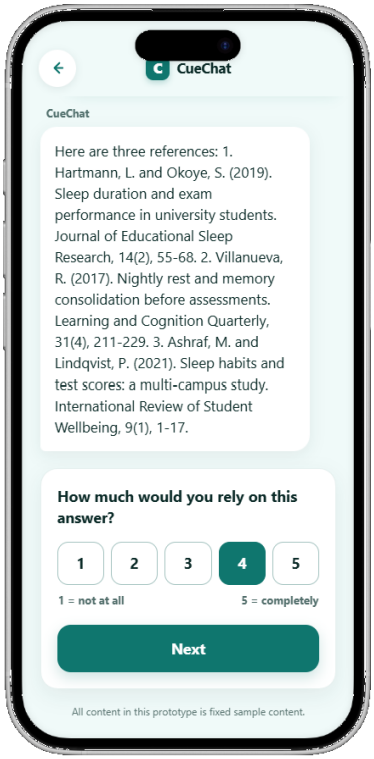
Question 1: the rating card



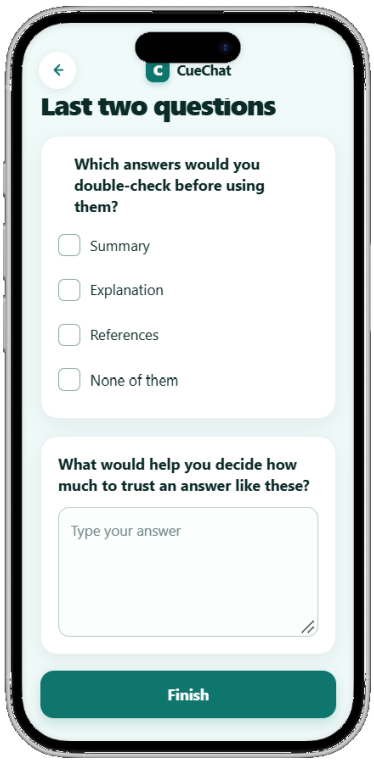
Question 2, explanation



Question 3, references

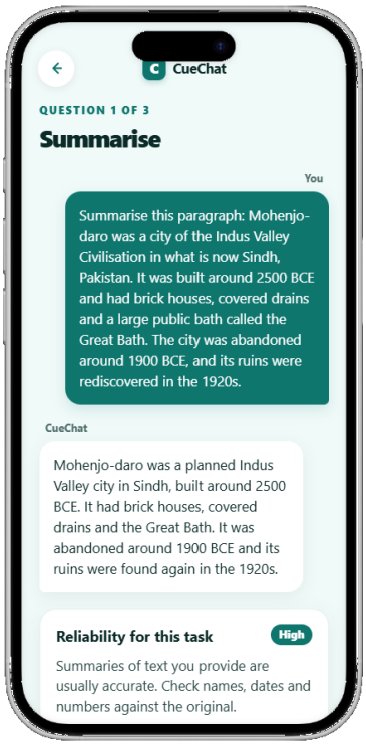


Question 3: the rating card

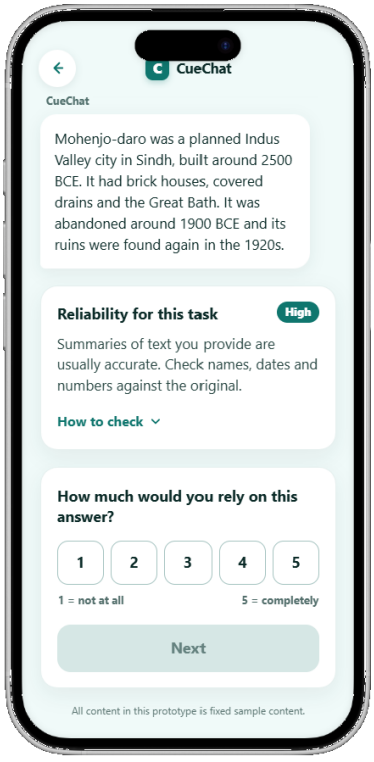


Final questions

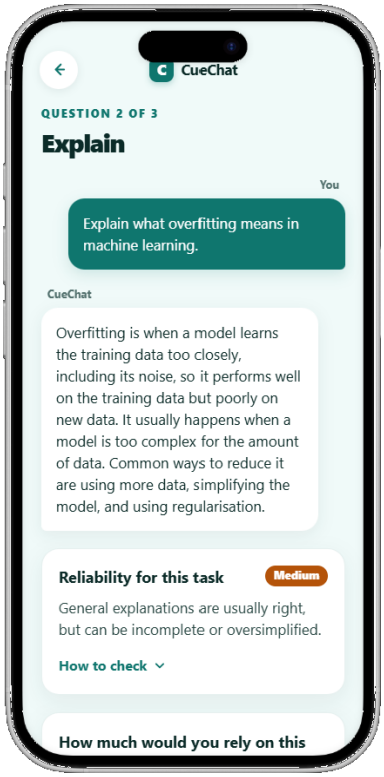
Version B (with reliability cues)



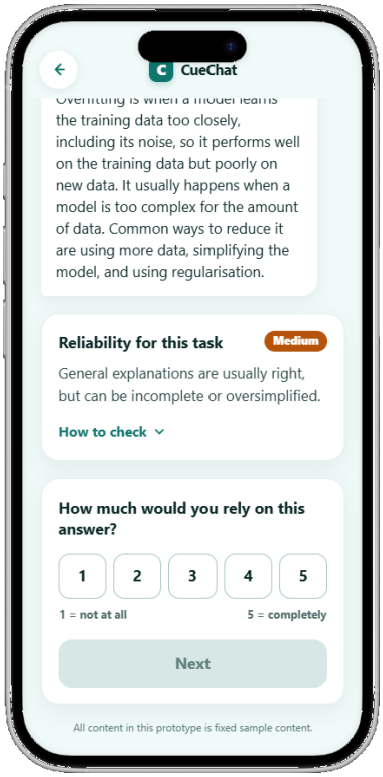
Question 1 with a High badge



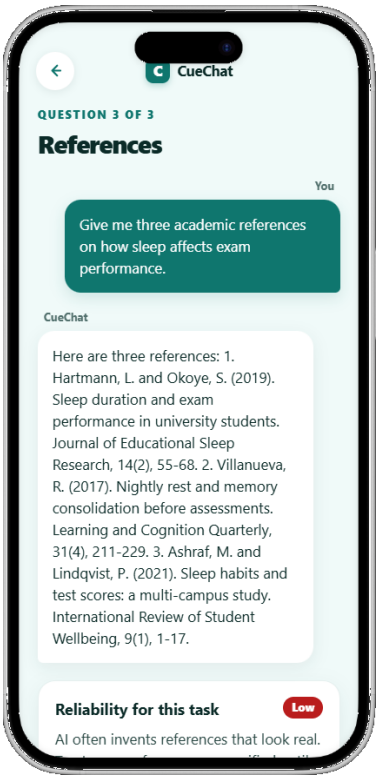
Question 1: reliability card and rating



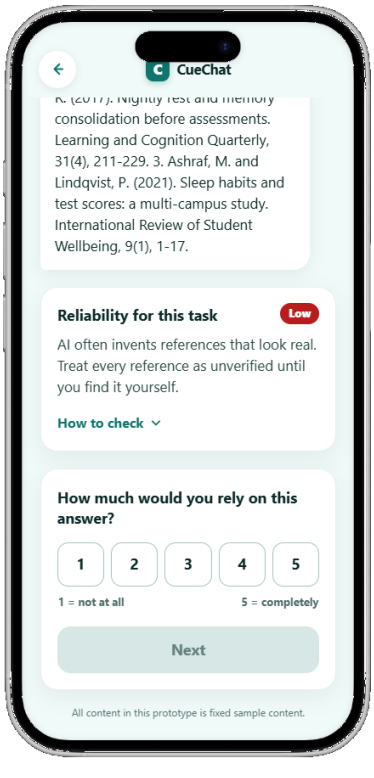
Question 2 with a Medium badge



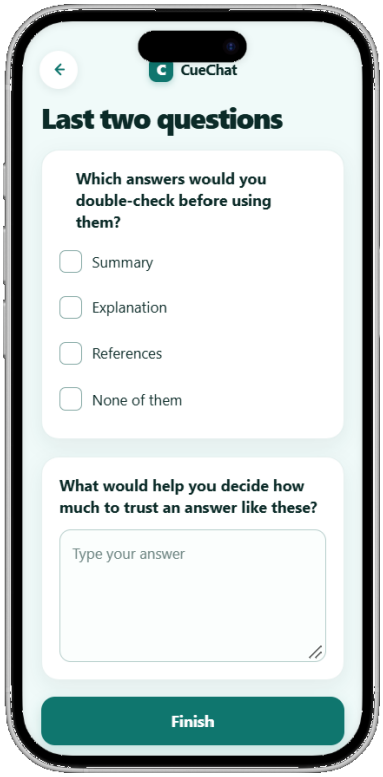
Question 2: reliability card and rating



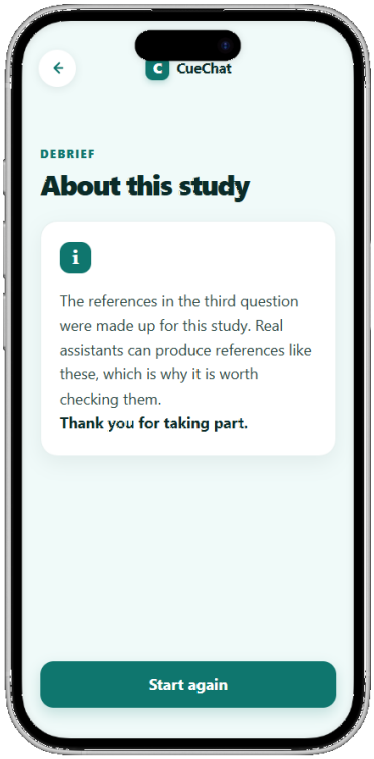
Question 3 with a Low badge



Question 3: reliability card and rating



Final questions (same in both versions)



Debrief (same in both versions)

In the Question 3 screenshot for Version A, the rating 4 is selected as a test tap and is not a participant's answer. The expanded “How to check” tips are not pictured.

4.4 User study and evaluation

CueChat was tried by three participants in a small, exploratory study. The aim was to see whether people's reliance ratings followed how reliable each kind of answer is, whether they chose to double-check the answer that was invented, and whether the reliability cards in Version B changed this. Three people cannot support statistical claims, so the findings below are observations.

Procedure

Each participant used one version only, one on Version A and two on Version B, because the references answer is the same in both and a second run would give it away. The planned instruction was to look at three questions asked to an AI assistant and its answers, and to say how much they would rely on each answer if it were real, without any mention of the reliability cards or of the references being invented. The exact wording used in each session was not recorded. After each answer they rated reliance from 1 to 5. At the end they ticked which answers they would double-check and wrote what would help them decide how much to trust an answer like these. The researcher noted which “How to check” tips were opened and whether the participant said they had noticed that the references were invented before the debrief.

*Table 1. Results from the three sessions. Sum., Expl., Refs: reliance ratings from 1 to 5 for the summary, explanation and references answers. *B1 first rated the references 2 and changed the rating to 1 after opening the tip.*

ID	Age	Ver.	Sum.	Expl.	Refs	Would double-check	Opened tips	Noticed refs invented
A1	20	A	3	4	4	Summary, explanation	Not applicable	No, thought they were correct
B1	21	B	5	3	1*	References	References	Expected them to be unreliable
B2	18	B	4	3	2	References	Summary, explanation	Yes

Findings

Reliance across tasks. In Version B both participants gave the references the lowest rating (1 and 2), against 5 and 4 for the summary, in the same order as the High, Medium and Low badges. In Version A, A1 gave the references 4, the same as the explanation and higher than the summary (3), and thought the references were correct. The gap between the summary and references ratings was 4 for B1 and 2 for B2, and −1 for A1.

What participants would double-check. Both Version B participants chose only the references. A1 chose the summary and the explanation and did not choose the references, which were the invented answer. This is the pattern the paper describes, where trust in references is not matched to how reliable they are [1], although it comes from one participant.

Role of the cards and tips. B1 expected the references to be unreliable because of the Low badge and lowered the rating from 2 to 1 after opening the tip for the references. B2 opened the tips for the summary and the explanation but not for the references, noticed that the references were invented, and said that this was why she rated them 2. Why B2 noticed it was not recorded, so it cannot be attributed to the Low badge.

Comments. All three asked for the sources to be visible. A1 wrote that sources should be quoted so people know where the references come from, and B2 asked for website or reference links along with the answer. B1 wrote that she trusted the reliability part but that “how to check” made everything useless, and said aloud that she would have used Google Scholar herself if she had wanted to, so she asked how the assistant saves her time. For B1, telling her to check the references herself removed the point of asking the assistant.

Possible changes

The sessions suggest showing where each reference or claim comes from, such as a link next to every reference, so the user is not left to search for it. A tip that only says to verify the answer yourself may be too costly for the task, so a version that links directly to the source might keep the benefit of the Low badge without making the assistant feel pointless. These changes were not built, and the study here tests the cards as they are.

Limitations

Three people aged 18 to 21 took part, one in Version A and two in Version B, so differences between the versions may reflect the people and not the cards, and one participant cannot stand for a version. Reliance was compared between participants, not within a person across versions, and all ratings are self-reported. The High, Medium and Low labels are our own judgement and the references were invented for the study, so the results say nothing about the accuracy of a real assistant. B1's rating was changed during the session and the later value is the one reported. Whether B2's noticing came from the cards was not recorded, and the exact instructions in each session were not recorded. The findings are descriptive, and a larger and balanced sample would be needed for any claim that reliability cues help trust follow reliability.

References

[1] K. Bouyzourn and A. Birch, “What shapes user trust in ChatGPT? A mixed-methods study of user attributes, trust dimensions, task context, and societal perceptions among university students,” arXiv:2507.05046, 2025.

[2] Y. Xu, S. Xiang, Y. Song, R. Sun, and X. Tong, “DuetUI: A bidirectional context loop for human-agent co-generation of task-oriented interfaces,” in Proc. CHI Conf. Human Factors in Computing Systems (CHI '26), Barcelona, Spain, 2026. arXiv:2509.13444.

[3] N. Abarca, A. Carvallo, C. López Moncada, and F. Bravo-Marquez, “Explaining AI without code: A user study on explainable AI,” in LatinX in AI Workshop at NeurIPS 2025. arXiv:2602.11159.

4.5 Final research report (IEEE format)

The final report for paper 4 follows in IEEE two-column format.

Task-Specific Reliability Cues in a Chat Assistant: A Prototype and Exploratory Study Inspired by Bouyzourn and Birch

Wardah Ahmed
Applied Human-Computer Interaction (CS 3014), Class 5-B, Student ID 24K-1045
National University of Computer and Emerging Sciences (FAST NUCES), Karachi, Pakistan

***Abstract*—People trust ChatGPT differently depending on the task, but chat interfaces present every answer in the same way. Bouyzourn and Birch found that trust was lowest for sources and called for explicit accuracy cues, without testing an interface. We built a clickable mock-up, CueChat, in two versions: Version A shows plain answers, and Version B adds a card under each answer stating how reliable that kind of answer usually is (High for a summary, Medium for an explanation, Low for references), with a tip on how to check it. The references answer was invented for the study. Three participants aged 18 to 21 each used one version and rated how much they would rely on each answer. In Version B (two participants) the references received the lowest rating and were the only answer chosen for double-checking. In Version A (one participant) the references were rated 4 and not chosen for checking. Participants asked for sources to be shown, and one said that a tip to check the references herself made the assistant feel pointless. The findings are exploratory.**

***Index Terms*—trust calibration, reliability cues, conversational agents, verification, prototype.**

I. INTRODUCTION

People rely on chat assistants for very different tasks, but the answers all arrive in the same form. Bouyzourn and Birch [1] studied what shapes trust in ChatGPT and found that it varies by task. They conclude that trust is calibrated by task verifiability, perceived objectivity and user expertise, and call for explicit accuracy cues. Their study tests no interface. This report asks whether a chat interface that shows task-specific reliability cues helps users match their reliance to how reliable the answer is.

II. RELATED WORK

Bouyzourn and Birch [1] surveyed 115 University of Edinburgh students and interviewed four. Perceived expertise was the strongest predictor of overall trust, and human-likeness did not predict it. Trust was highest for summarising (3.05), explaining (3.02) and coding (3.01) and lowest for sources (2.19). Yet overall trust rose with confidence in generating references, which the authors read as automation bias, persuasive fluency mistaken for factual reliability. Interviewees described trust dropping after they found references that did not exist.

Other work in this set points the same way. Abarca et al. [2] found that novices trusted an explanation module more readily than experts, and Xu et al. [3] reported a participant whose trust in a generated interface collapsed after incorrect information. Neither tests whether a cue shown before an answer is used helps users set their trust appropriately.

III. RESEARCH GAP AND QUESTION

The authors of [1] call for explicit accuracy cues and user education but do not test whether an interface can help users bring their trust in line with reliability for each task, and their cross-sectional, self-reported design cannot show whether cues change trust. Our research question is: does a chat interface that shows task-specific reliability cues help users bring their reliance in line with how reliable the answer is, compared with the same interface without the cues?

IV. PROTOTYPE

CueChat is a clickable mobile mock-up built in Figma Make. It is inspired by [1] and is not a real assistant. All content is fixed sample content. Every participant sees three tasks in the same order. The first asks for a summary of a short paragraph about Mohenjo-daro, and the summary is accurate. The second asks for an explanation of overfitting, and the explanation is correct. The third asks for three academic references on sleep and exam performance, and the references were invented for the study. After each answer the user rates how much they would rely on it, from 1 to 5.

Version A shows the answer and the rating question. Version B adds a “Reliability for this task” card under each answer: High for the summary, Medium for the explanation and Low for the references, with a one-line reason and a “How to check” tip. The levels follow the task ordering reported in [1] and are our own judgement, not a measured accuracy.

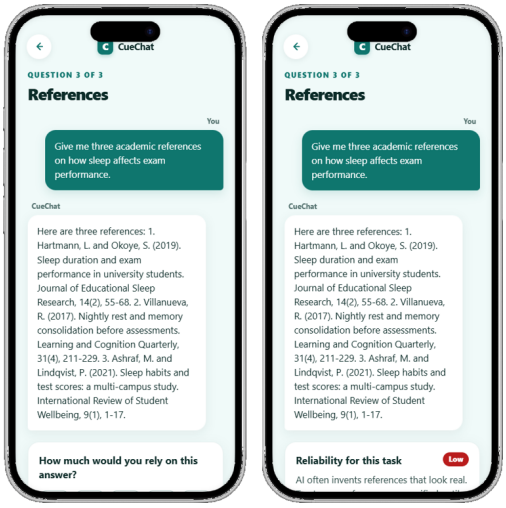


Fig. 1. The references question in Version A (left) and Version B (right).

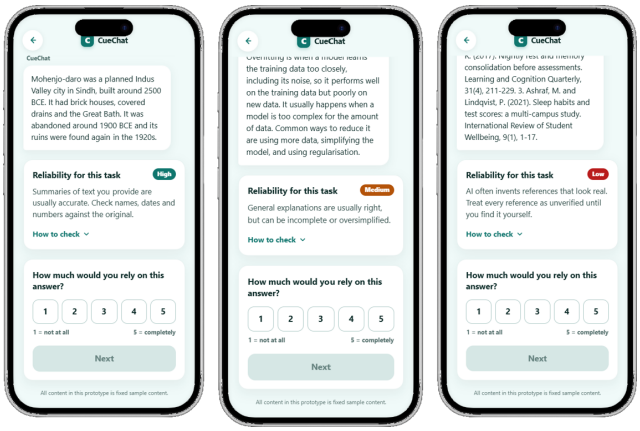


Fig. 2. The reliability cards in Version B: High, Medium and Low.

At the end the user ticks which answers they would double-check and writes what would help them decide how much to trust an answer like these. A debrief screen states that the references were made up.

V. EVALUATION

Three people aged 18 to 21 took part, one in Version A and two in Version B. Each used one version only, because the references answer is the same in both. The planned instruction was to look at the three answers and say how much they would rely on each if it were real, with no mention of the cards or of the references being invented. The researcher noted which tips were opened and whether the participant said they had noticed that the references were invented. The exact wording in each session was not recorded.

VI. RESULTS

TABLE I. Reliance ratings (1 to 5) for the summary, explanation and references, and the answers chosen for double-checking. *B1 first rated 2 and changed to 1 after opening the tip.

ID	Ver.	Sum.	Expl.	Refs	Would check
A1	A	3	4	4	Sum., expl.
B1	B	5	3	1*	Refs
B2	B	4	3	2	Refs

In Version B both participants rated the references lowest (1 and 2), in the same order as the badges, and both chose only the references to double-check. In Version A, A1 rated the references 4, thought they were correct, and chose the summary and the explanation instead. The gap between the summary and references ratings was 4 for B1, 2 for B2 and –1 for A1. B1 expected the references to be unreliable because of the Low badge and lowered her rating after opening the tip. B2 opened the tips for the other two answers, noticed that the references were invented, and said this was why she rated them 2. Why B2 noticed was not recorded.

All three asked for sources to be shown, for example quoted or as links. B1 wrote that she trusted the reliability card but that the check tip made everything useless, and said that if she wanted to use Google Scholar she would have, so she asked how the assistant saves her time.

VII. DISCUSSION

The pattern matches what the cards were meant to do: with them, participants rated and checked the unreliable answer differently from the reliable ones, and without them one participant trusted invented references. With one participant in Version A and two in Version B, this cannot be attributed to the cards, since B2 may have noticed the invented references without them. The comments are more informative than the ratings. Participants wanted to see where an answer comes from, and a tip that tells the user to verify the answer alone can remove the point of using the assistant. A card that links to the source might keep the benefit of the Low badge without that cost. This is a suggestion from the sessions and was not tested.

VIII. LIMITATIONS AND FUTURE WORK

Three participants aged 18 to 21 took part, one in Version A, so differences may reflect the people. Reliance was compared between participants, ratings are self-reported, and the labels are our own judgement. The references were invented, so nothing here describes a real assistant's accuracy. B1's rating changed during the session, and the exact instructions were not recorded. Future work should use a balanced and larger sample, record whether participants notice the invented references, and test a card that links to sources.

IX. CONCLUSION

We built two versions of a chat mock-up to test whether task-specific reliability cues help users match their reliance to reliability. With three participants, those who saw the cards rated and checked the invented references differently from the other answers, and the participant without them trusted the references. The sample is too small to show that the cards caused this. Participants wanted visible sources, and a larger study should test cards that link to them.

REFERENCES

[1] K. Bouyzourn and A. Birch, “What shapes user trust in ChatGPT? A mixed-methods study of user attributes, trust dimensions, task context, and societal perceptions among university students,” arXiv:2507.05046, 2025.

[2] N. Abarca, A. Carvallo, C. López Moncada, and F. Bravo-Marquez, “Explaining AI without code: A user study on explainable AI,” in *LatinX in AI Workshop at NeurIPS 2025*, arXiv:2602.11159.

[3] Y. Xu, S. Xiang, Y. Song, R. Sun, and X. Tong, “DuetUI: A bidirectional context loop for human-agent co-generation of task-oriented interfaces,” in *Proc. CHI Conf. Human Factors in Computing Systems (CHI '26)*, Barcelona, Spain, 2026, arXiv:2509.13444.

Paper 5: Explaining AI Without Code

Abarca, Carvallo, López Moncada, and Bravo-Marquez. LatinX in AI Workshop at NeurIPS 2025. arXiv:2602.11159.

5.1 Literature review

As AI tools reach people who do not write code, explaining a model's predictions to non-technical users has become a design problem as much as a technical one. Abarca et al. [1] work on this inside DashAI, an open-source no-code machine learning platform, by adding an explainability module for tabular classification. The module combines three explanation methods: Partial Dependence Plots (PDP), Permutation Feature Importance (PFI), and KernelSHAP.

The authors ran a between-subjects user study with 20 participants, 10 novices and 10 experts. Participants were split across two scenarios, with 5 novices and 5 experts in each. In Scenario A the explanations were generated and analyzed one after another (PDP, then PFI, then KernelSHAP). In Scenario B all three were generated and analyzed together. They measured task success across five explainability tasks, satisfaction with the explanations using the Explanation Satisfaction Scale (taken from Hoffman et al., 2018), which covers comprehension, satisfaction, sufficiency of details, completeness, usability, usefulness, accuracy, and trust, and four subscales of the Trust in Automation questionnaire: familiarity, predictability, propensity to trust, and confidence in automation [1].

All five tasks reached at least 80 percent success, and most errors came in the last three tasks (PDP, KernelSHAP, and comparing the three methods). Satisfaction medians were 4 or 5, with usefulness and accuracy highest. There was no significant difference between the progressive and combined scenarios ($U = 38$, $p = 0.38$), so the order of presentation did not change how satisfied people were. Novices rated the explanations more positively than experts, though the effect was only marginally significant ($p = 0.082$), and experts were more critical of sufficiency of details and completeness. On trust, novices trusted the system more readily than experts, while predictability and confidence showed moderate to high agreement, which the authors read as the explanations helping users judge when to rely on the model [1].

The authors state several limits. The sample was small ($N = 20$), the reliability of the trust questionnaire was questionable (Cronbach's alpha of 0.60), the novice-expert effect was only marginal, and the analysis covered only the five explainability tasks. They recommend adaptive explanation strategies that adjust depth and interpretability to user expertise and task context, and plan to extend the work to explanations for large language models [1]. The adaptive strategies were recommended, not built or tested.

The pattern of novices and experts judging the same system differently appears elsewhere in this set. Bouyzourn and Birch [2] found that trust in ChatGPT rose with frequent use and fell with a better understanding of how language models work, which fits experts here being more critical and novices trusting more readily. Xu et al. [3] found that most participants felt DuetUI lowered the barrier to entry, while one cautioned that older adults might still struggle.

5.2 Problem identification and research gap

Problem identification

Novices and experts judged the same explanations differently. Novices found them useful and trustworthy, while experts wanted more detail and completeness. A single fixed set of explanations has to choose a depth, and either choice short-changes one group: too little for experts, or possibly too much for novices. Changing the order of presentation made no significant difference, so sequencing is not the lever. What the results point to is the user's level of expertise. A layered design, which Shneiderman lists as his fourth criterion for direct manipulation [4], would let a beginner start with a simple view while an expert opens more detail.

Research gap

The authors found that novices and experts respond differently to the same explanations, found that presentation order did not change satisfaction, and recommend adaptive explanations that adjust depth to user expertise [1]. There is no evidence on whether an explanation interface that starts simple and offers more detail on request improves satisfaction for both groups. The study also had only 20 participants, with 5 novices and 5 experts per scenario, so its group differences are tentative. This project tests the layered design.

Research question

Does an explanation interface that starts with a simple view and lets users open more detail lead to higher satisfaction among both novices and experts than a single fixed set of explanations? The prototype in section 5.3 is built to answer this.

5.3 Prototype

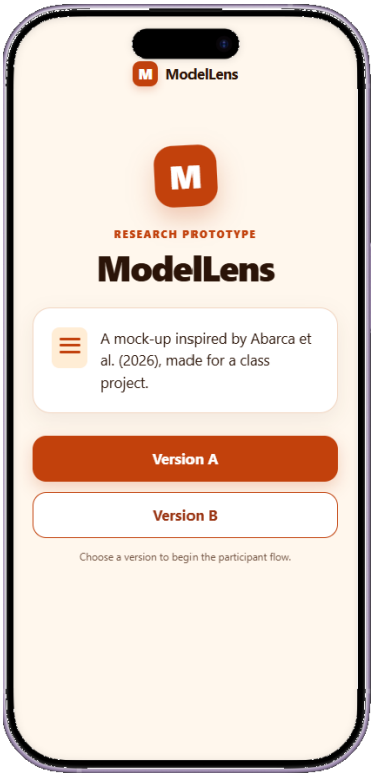
The prototype, ModelLens, is a clickable mobile mock-up inspired by the explainability module of Abarca et al. [1], built in Figma Make. It is not the authors' system and no real model is connected. A sample no-code tool predicts that a student, Student 14, will not pass a course (72 percent likely), and the user is shown an explanation. The student's data are fixed sample content, which is stated on screen. The explanation uses two of the kinds of explanation in [1], a permutation feature importance chart and a partial dependence chart, in a simplified form. KernelSHAP is not included. Prototype link: Figma Make prototype.

The prototype has two versions, and each participant sees only one. Version A shows one fixed explanation: the importance chart, the partial dependence chart and a comparison of the student with the average, all at once and all technical. Version B contains exactly the same content in three layers. Level 1 gives a plain-language summary (low attendance and few study hours) with two highlighted facts, Level 2 opens the importance chart under the heading “How much each factor matters”, and Level 3 opens the technical detail, including the partial dependence chart. The user opens each level with a button, so a beginner can stop at Level 1 and an expert can go on. Because the content is the same in both versions, any difference comes from how it is organised.

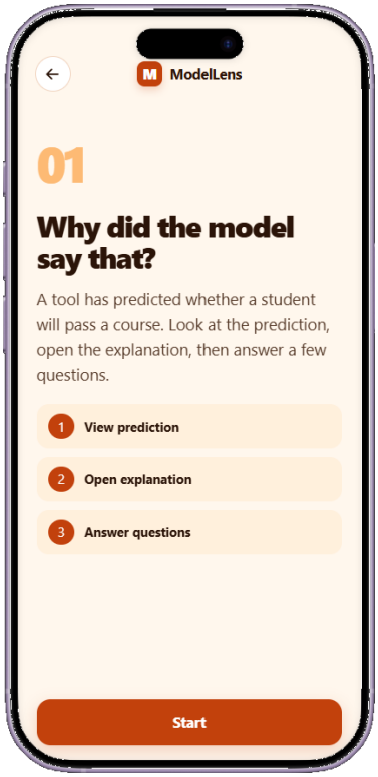
After the explanation, the user answers a check question (which factor mattered most, where the correct answer, Attendance, is the longest bar), rates four statements from 1 to 5 (I understand why the model made this prediction; the explanation gave me enough detail; it was easy to follow; I would trust the prediction), rates their own familiarity with machine learning

from 1 to 5, and says what they would change. The familiarity rating is used to separate novices from experts. These are four single items written for this project, in the spirit of the satisfaction and trust measures in [1], and are not the validated scales used by the authors.

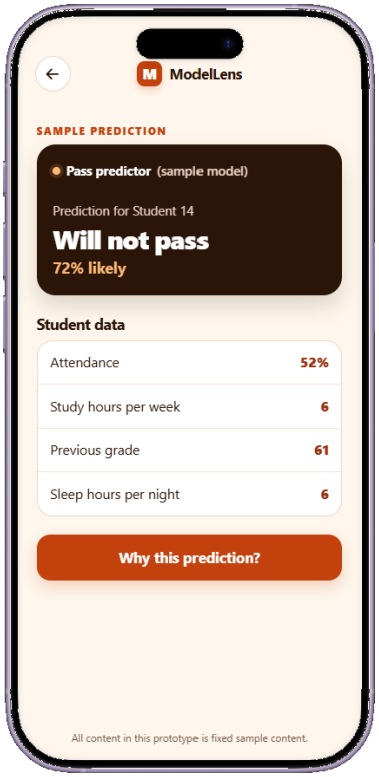
Screens shared by both versions



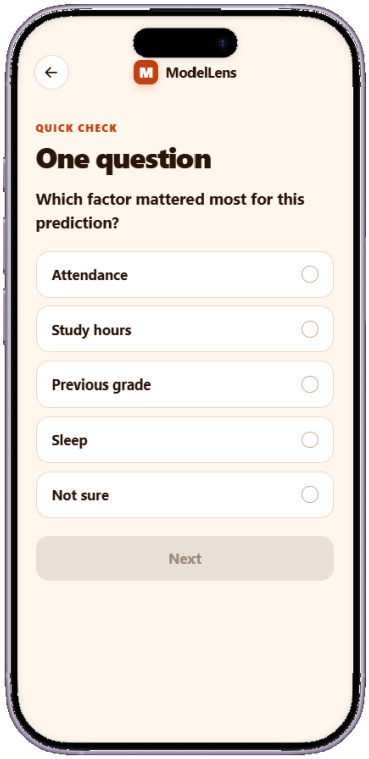
Start screen with the two versions



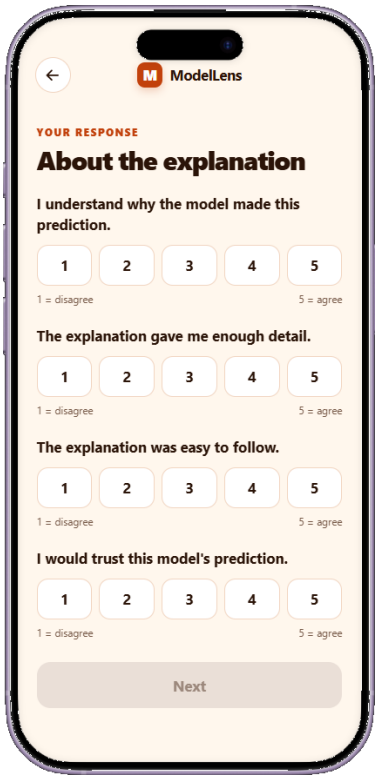
Introduction



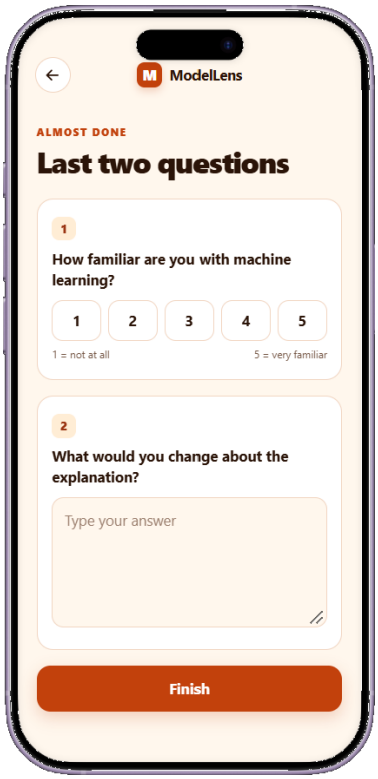
Prediction for Student 14



Check question

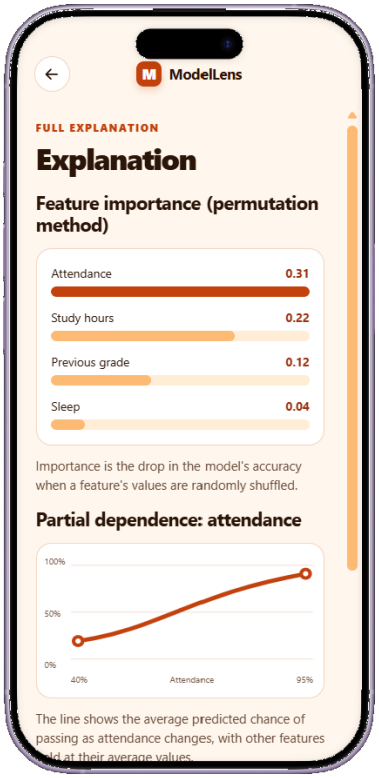


Ratings of the explanation

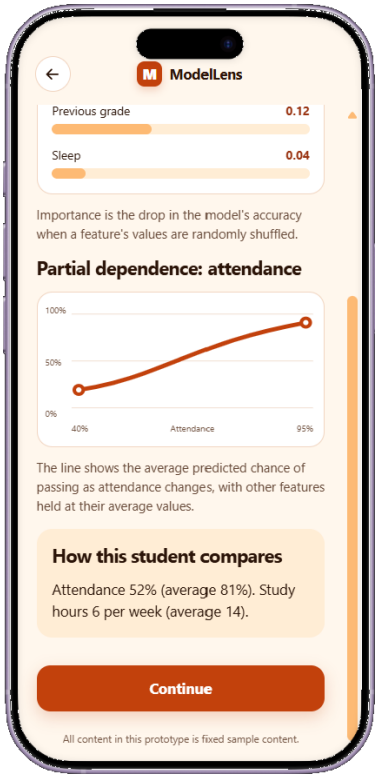


Familiarity and comments

Version A (one fixed explanation)

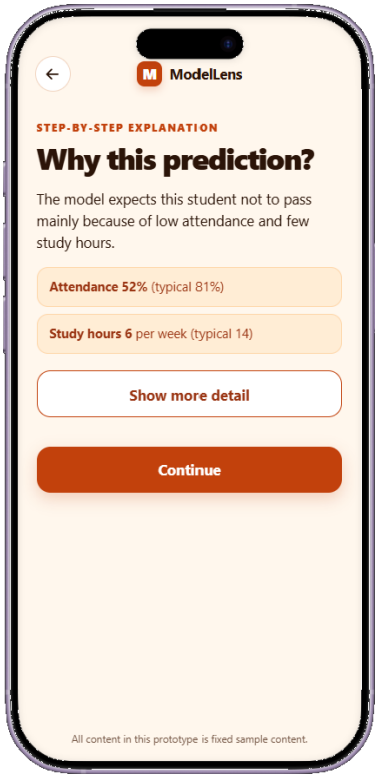


Explanation: importance chart and partial dependence

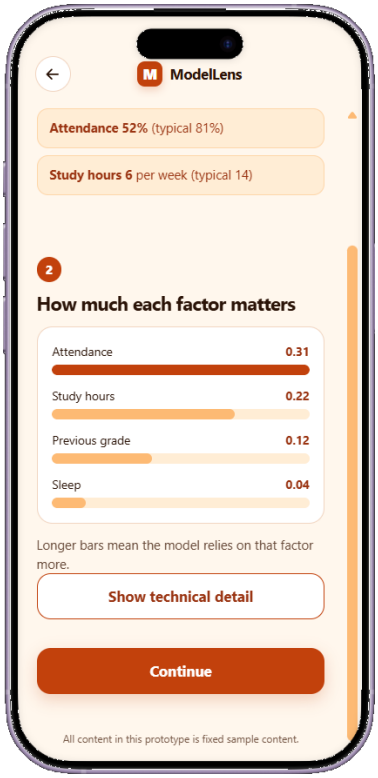


Explanation: comparison and Continue

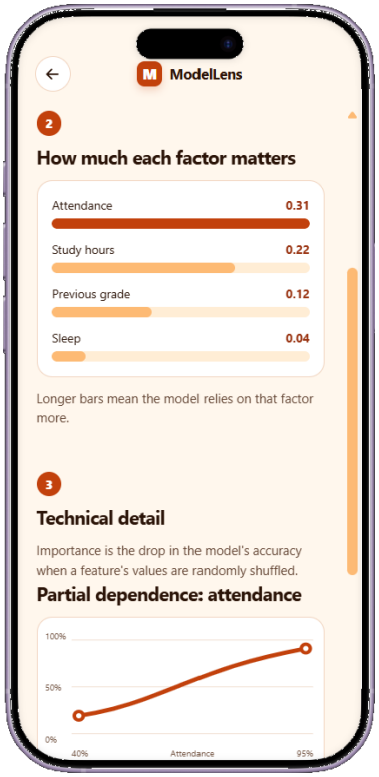
Version B (three layers)



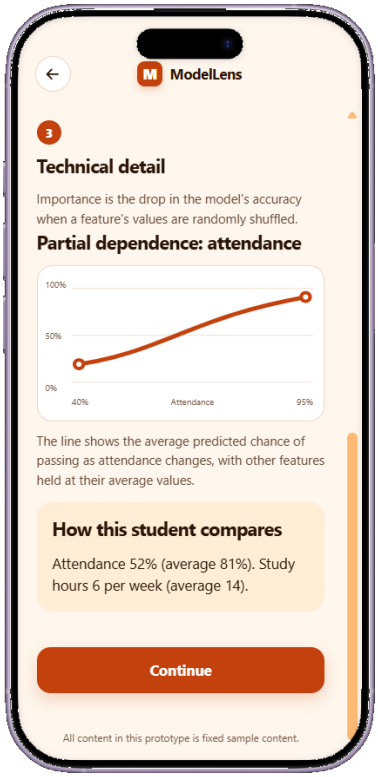
Level 1: plain-language summary



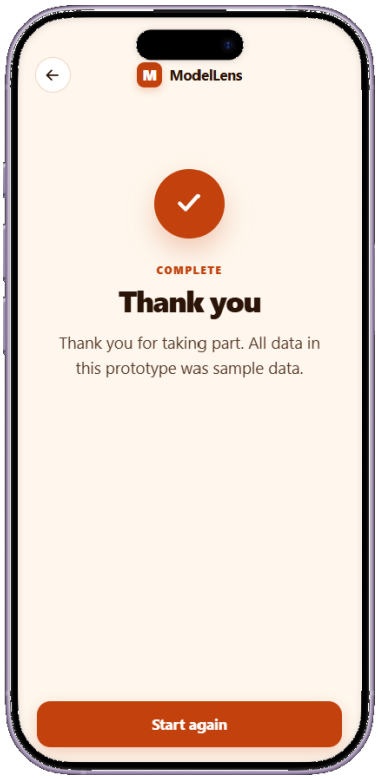
Levels 1 and 2: how much each factor matters



Level 3: technical detail opens



Level 3: comparison and Continue



Thank you screen (same in both versions)

5.4 User study and evaluation

ModelLens was tried by three participants in a small, exploratory study. The aim was to see how people with different familiarity with machine learning rated the explanation, whether they could say which factor mattered most, and how the layered version was used. Three people cannot support statistical claims, so the findings below are observations.

Procedure

Each participant used one version only, two on Version A and one on Version B. The planned instruction was to look at the prediction, open the explanation and answer a few questions, with no mention of layers or of there being two versions. The exact wording used in each session was not recorded. After the explanation the participant answered the check question, rated four statements from 1 to 5 (understanding, enough detail, ease of following, trust in the prediction), rated their own familiarity with machine learning from 1 to 5, and could write what they would change. For Version B the researcher noted which levels were opened. For the comparison, a familiarity rating of 1 is treated as novice and 4 or 5 as familiar with machine learning. No participant rated 2 or 3.

Table 1. Results from the three sessions. Fam.: self-rated familiarity with machine learning (1 to 5). Underst., Detail, Easy, Trust: agreement (1 to 5) that the participant understood why the model made the prediction, that the explanation gave enough detail, that it was easy to follow, and that they would trust the prediction.

ID	Age	Ver.	Fam.	Check answer	Underst.	Detail	Easy	Trust
A1	18	A	1	Attendance	4	2	4	2
A2	23	A	5	Attendance	4	3	3	3
B1	22	B	4	Attendance	5	4	5	2

Findings

Check question. All three participants answered Attendance, the factor with the longest bar, so everyone could read the importance chart in both versions.

Ratings. In Version A (two participants) the average ratings were 4.0 for understanding, 2.5 for enough detail, 3.5 for ease and 2.5 for trust. B1, the only Version B participant, gave 5, 4, 5 and 2, opened all three levels, and so saw everything that Version A shows. B1 gave the highest ratings for understanding, detail and ease, but trust was no higher than A1's. With one participant in Version B, who was also familiar with machine learning, this cannot be read as an effect of the layers.

Novice and familiar users. In Version A, the novice (A1) rated the explanation easier to follow than the familiar participant (A2) did (4 against 3), and rated detail lower (2 against 3) and trust lower (2 against 3). The paper found novices trusting the explanations more readily than experts [1]. This pair does not show that pattern, though two people cannot contradict it.

Comments. A2 wrote nothing. A1 wrote that there is no way to know the personal circumstances behind the 72 percent chance of failing. B1 wrote that the model should have some diagnostic of the student's intellect, since smarter students can do well with low attendance and few hours of study. Both comments are about what the model leaves out, not about how deep or how organised the explanation was. B1 rated the explanation clear (5 for understanding) and still gave a trust rating of 2, which suggests that for B1 the limit on trust was the model and not the explanation.

Possible changes

Both comments suggest adding a note about what the model does not know, such as personal circumstances or ability, so that users can see the limits of the prediction. This was not built. The layered design itself was not tested in a fair comparison, and the next step would be to run novices and familiar users in both versions.

Limitations

Three people aged 18 to 23 took part, two in Version A and one in Version B, and there was no novice in Version B, so the groups are not matched and the study cannot compare the versions. All ratings are self-reported single items written for this project, not the validated satisfaction and trust scales used in [1]. The explanations are simplified, use fixed sample data, and do not include KernelSHAP. One participant wrote no comment, and the exact instructions in each session were not recorded. The findings are descriptive, and a larger sample with novices and familiar users in both versions would be needed for any claim about layered explanations.

References

[1] N. Abarca, A. Carvallo, C. López Moncada, and F. Bravo-Marquez, “Explaining AI without code: A user study on explainable AI,” in LatinX in AI Workshop at NeurIPS 2025. arXiv:2602.11159.

[2] K. Bouyzourn and A. Birch, “What shapes user trust in ChatGPT? A mixed-methods study of user attributes, trust dimensions, task context, and societal perceptions among university students,” arXiv:2507.05046, 2025.

[3] Y. Xu, S. Xiang, Y. Song, R. Sun, and X. Tong, “DuetUI: A bidirectional context loop for human-agent co-generation of task-oriented interfaces,” in Proc. CHI Conf. Human Factors in Computing Systems (CHI '26), Barcelona, Spain, 2026. arXiv:2509.13444.

[4] B. Shneiderman, “Direct manipulation: A step beyond programming languages,” Computer, vol. 16, no. 8, pp. 57-69, 1983.

5.5 Final research report (IEEE format)

The final report for paper 5 follows in IEEE two-column format.

Layered Explanations for a No-Code Machine Learning Tool: A Prototype and Exploratory Study Inspired by Abarca et al.

Wardah Ahmed
Applied Human-Computer Interaction (CS 3014), Class 5-B, Student ID 24K-1045
National University of Computer and Emerging Sciences (FAST NUCES), Karachi, Pakistan

***Abstract*—Explaining a model's predictions to non-technical users is a design problem. Abarca et al. added explanations to a no-code machine learning platform and found that novices and experts judged them differently, and they recommend adaptive explanations that match depth to the user's expertise without testing them. We built a clickable mock-up, ModelLens, in which a sample model predicts that a student will not pass. Version A shows one fixed technical explanation. Version B shows the same content in three layers, from a plain-language summary to technical detail, opened on request. Three participants aged 18 to 23 took part, two on Version A and one on Version B. All three identified the most important factor correctly. In Version A the novice found the explanation easier to follow but less detailed and less trustworthy than the participant familiar with machine learning. The Version B participant rated understanding and ease highest but trusted the prediction little. Two participants commented on what the model leaves out, not on the explanation. The study cannot compare the versions, and the findings are exploratory.**

***Index Terms*—explainable AI, no-code machine learning, layered explanations, novice and expert users, prototype.**

I. INTRODUCTION

As machine learning tools reach people who do not write code, explaining a model's predictions has become a design problem as much as a technical one. Abarca et al. [1] address it inside DashAI, an open-source no-code platform, with an explainability module for tabular classification. They found that novices and experts reacted differently to the same explanations and recommend adaptive strategies that adjust depth to expertise, but they did not build or test them. This report asks whether an explanation that starts simple and opens more detail on request works for both groups.

II. RELATED WORK

The module in [1] combines Partial Dependence Plots, Permutation Feature Importance and KernelSHAP. In a between-subjects study with 20 participants, 10 novices and 10 experts, all five tasks reached at least 80 percent success and satisfaction medians were 4 or 5. Presenting the explanations one after another or all together made no significant difference. Novices rated the explanations somewhat more positively than experts, though only marginally significantly, and experts were more critical of detail and completeness. Novices also trusted the system more readily.

A layered design, which Shneiderman lists as a criterion for direct manipulation [4], would let a beginner start with a simple view while an expert opens more detail. The same difference between novices and experts appears in other work in this set. Bouyzourn and Birch [2] found that trust in ChatGPT rose with frequent use and fell with better understanding of how language models work.

III. RESEARCH GAP AND QUESTION

The authors of [1] found that novices and experts respond differently, found that presentation order did not change satisfaction, and recommend adaptive explanations, but there is no evidence on whether an interface that starts simple and offers detail on request improves satisfaction for both groups. Our research question is: does an explanation interface that starts with a simple view and lets users open more detail lead to higher satisfaction among both novices and experts than a single fixed set of explanations?

IV. PROTOTYPE

ModelLens is a clickable mobile mock-up built in Figma Make, inspired by the module in [1]. It is not the authors' system, and no real model is connected. A sample pass predictor says that Student 14 will not pass (72 percent likely), and the user is shown an explanation made of a permutation importance chart, a partial dependence chart and a comparison with average values. KernelSHAP is not included, and all values are fixed sample data.

Version A shows all of this at once in a single technical screen. Version B contains exactly the same content in three layers: a plain-language summary with two highlighted facts, then the importance chart under the heading “How much each factor matters”, then the technical detail including the partial dependence chart. Each layer opens with a button. Because the content is identical, differences come from how it is organised.

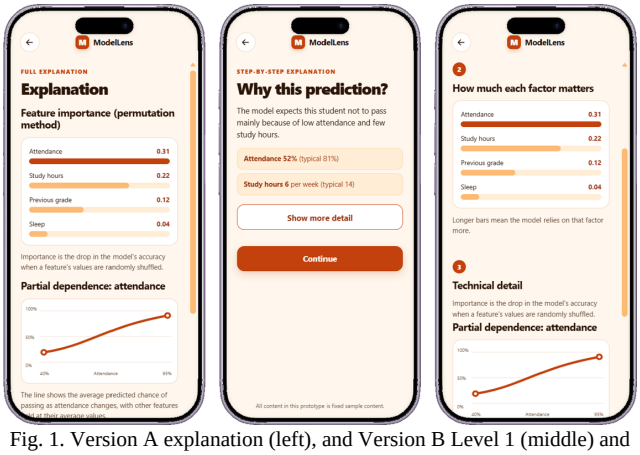


Fig. 1. Version A explanation (left), and Version B Level 1 (middle) and Level 3 (right).

After the explanation the user answers a check question, rates four statements from 1 to 5 (understanding, enough

detail, ease of following, trust in the prediction), rates their own familiarity with machine learning, and can write what they would change. These are four single items written for this project and are not the validated scales used in [1].

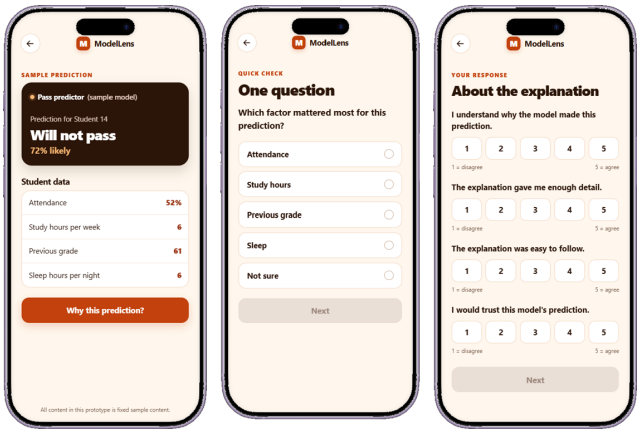


Fig. 2. The prediction screen, the check question and the rating screen.

V. EVALUATION

Three people aged 18 to 23 took part, two in Version A and one in Version B, each using one version. The planned instruction was to look at the prediction, open the explanation and answer a few questions, without mentioning layers or two versions. The exact wording in each session was not recorded. For Version B the researcher noted which levels were opened. A familiarity rating of 1 is treated as novice and 4 or 5 as familiar with machine learning.

VI. RESULTS

TABLE I. Ratings from 1 to 5. Fam.: familiarity with machine learning. Und.: understood why the model made the prediction. Det.: enough detail. Easy: easy to follow. Trust: would trust the prediction.

ID	Ver.	Fam.	Und.	Det.	Easy	Trust
A1	A	1	4	2	4	2
A2	A	5	4	3	3	3
B1	B	4	5	4	5	2

All three participants answered the check question correctly, with Attendance. In Version A the average ratings were 4.0 for understanding, 2.5 for detail, 3.5 for ease and 2.5 for trust. The novice (A1) found the explanation easier to follow than the familiar participant (A2) did (4 against 3) but rated detail (2 against 3) and trust (2 against 3) lower. B1 opened all three levels, gave the highest ratings for understanding (5), detail (4) and ease (5), and rated trust 2.

A2 wrote no comment. A1 wrote that there is no way to know the personal circumstances behind the 72 percent chance of failing. B1 wrote that the model should have some diagnostic of the student's intellect, since smarter students can do well with low attendance and few hours of study.

VII. DISCUSSION

The study cannot answer the research question. Version B had one participant, who was familiar with machine learning and opened every level, so his higher ratings may come from the layers or from his familiarity. The paper found novices trusting explanations more than experts [1], and the pair in Version A does not show that, although two people cannot contradict it. The comments are informative: both were about what the model leaves out, not about how the explanation was organised. B1 understood the explanation and still gave trust a 2, which suggests that for him the limit on trust was the model and not the explanation. A note on what the model does not know, such as personal circumstances or ability, is a possible addition that was not tested.

VIII. LIMITATIONS AND FUTURE WORK

Three participants aged 18 to 23 took part, with no novice in Version B, so the groups are not matched. Ratings are self-reported single items, the explanations are simplified and use fixed sample data without KernelSHAP, one participant left no comment, and the instructions were not recorded. Future work should run novices and familiar users in both versions with a larger sample and the validated scales from [1], and record which layers each participant opens.

IX. CONCLUSION

We built two versions of an explanation screen, one fixed and one layered, to test whether starting simple and opening detail on request works for both novices and experts. With three participants the versions could not be compared. Participants could read the importance chart, and their comments pointed to the limits of the model rather than the explanation. A larger and balanced study is needed to test layering.

REFERENCES

[1] N. Abarca, A. Carvallo, C. López Moncada, and F. Bravo-Marquez, “Explaining AI without code: A user study on explainable AI,” in *LatinX in AI Workshop at NeurIPS 2025*, arXiv:2602.11159.

[2] K. Bouyzourn and A. Birch, “What shapes user trust in ChatGPT? A mixed-methods study of user attributes, trust dimensions, task context, and societal perceptions among university students,” arXiv:2507.05046, 2025.

[3] Y. Xu, S. Xiang, Y. Song, R. Sun, and X. Tong, “DuetUI: A bidirectional context loop for human-agent co-generation of task-oriented interfaces,” in *Proc. CHI Conf. Human Factors in Computing Systems (CHI '26)*, Barcelona, Spain, 2026, arXiv:2509.13444.

[4] B. Shneiderman, “Direct manipulation: A step beyond programming languages,” *Computer*, vol. 16, no. 8, pp. 57-69, 1983.